Power Electronics

Previous Year Questions • 1991–2026

GATE ELECTRICAL ENGINEERING

(EE)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Power Electronics (EE), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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- AIR 27 — GATE (ECE)
- AIR 45 — GATE (IN)
- 5+ years of teaching experience

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- AIR 19 — GATE (IN)
- AIR 66 — GATE (EE)
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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

Easy Medium Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC **Questions:** 45 **MCQ:** 30 **MSQ:** 3 **NAT:** 12
1M: 20 **2M:** 22 **Descriptive:** 3 **Avg Weightage** ~4M **Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

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UNIT

I

Power Semiconductor Devices

Topics

1. Power Diode
2. Power Transistor
3. Static V-I Characteristics of Thyristor (SCR)
4. Static V-I Characteristics of MOSFET and IGBT
5. Firing and Gating Circuits for Thyristor, MOSFET and IGBT
6. Protection Schemes, Series and Parallel Operation of SCR
7. V-I Characteristics of GTO, TRIAC and Switch Modules

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Unit Snapshot*

Stream: EE	**Questions:** 53	**MCQ:** 43	**MSQ:** 1	**NAT:** 9
1M: 37	**2M:** 16	**Descriptive:** 0	**Avg Weightage** ~2M	**Range:** 1M(2024)–5M(2010)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 MCQ 2026 1M

The figure shows a straight-line approximation for the forward characteristics of a power diode. A continuous on-state current of 15 A is flowing through the diode.

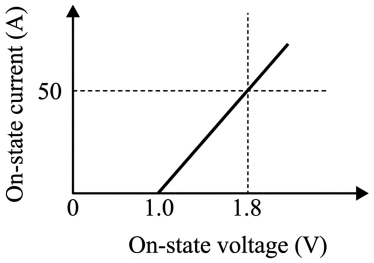


Fig. Q1.26.1

What is the power loss in the diode?

A. 32.8 W **B.** 21.2 W **C.** 18.6 W **D.** 23.1 W

Q1.26.2 MCQ 2026 1M

Consider the circuit shown in Figure (a). A gate pulse V_g is applied between time instants t_0 and t_1. After t_1, during the MOSFET turn-OFF process, it experiences a voltage overshoot.

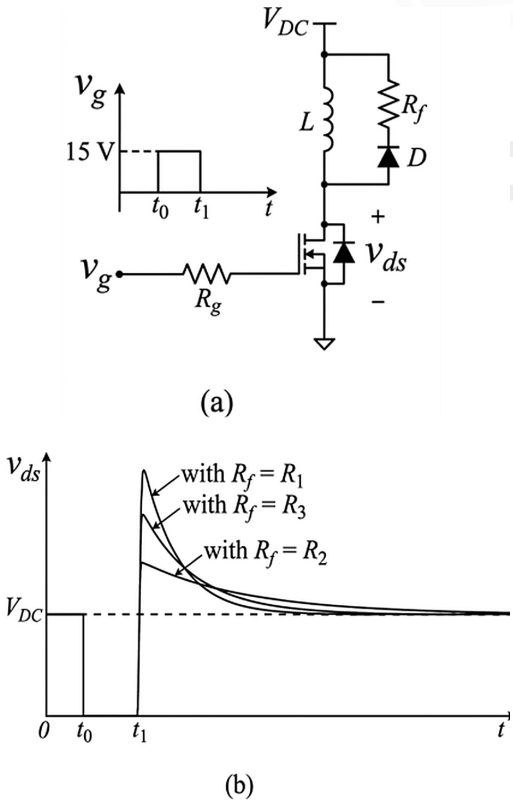


Fig. Q1.26.2

Based on the V_{ds} waveforms shown in Figure (b), which one of the following options is correct?

A. $R_1 > R_2 > R_3$ **B.** $R_1 > R_3 > R_2$
C. $R_3 > R_2 > R_1$ **D.** $R_2 > R_3 > R_1$

Q1.24.1 MCQ 2024 1M

If the following switching devices have similar power ratings, which one of them is the fastest?

A. SCR **B.** GTO **C.** IGBT **D.** Power MOSFET

Q1.23.1 MSQ 2023 1M

A semiconductor switch needs to block voltage V of only one polarity ($V > 0$) during the OFF state as shown in Figure (i) and carry current in both directions during the ON state as shown in Figure (ii). Which of the following switch combination(s) will realize the same?

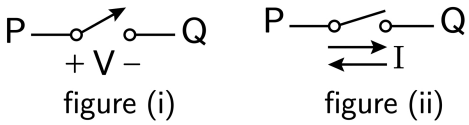
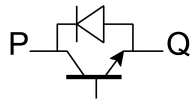
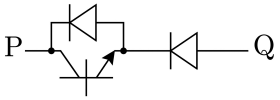
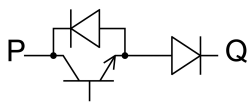
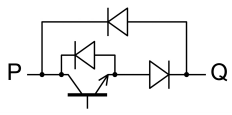


Fig. Q1.23.1

A. 

B. 

C. 

D. 

Q1.23.2 NAT 2023 1M

The circuit shown in the figure has reached steady state with thyristor T in the OFF condition. Assume that the latching and holding currents of the thyristor are zero. The thyristor is turned ON at $t = 0$ s. The duration in microseconds for which the thyristor would conduct before it turns OFF is _____ (Round off to two decimal places).

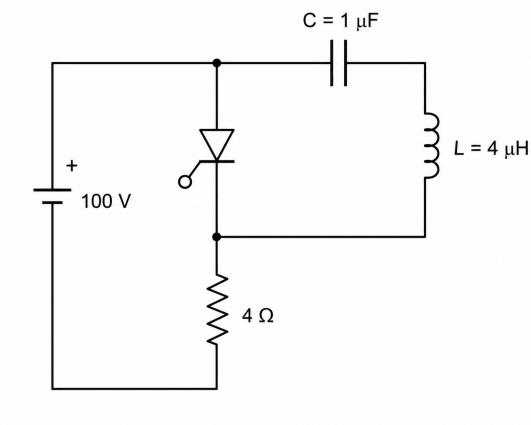


Fig. Q1.23.2

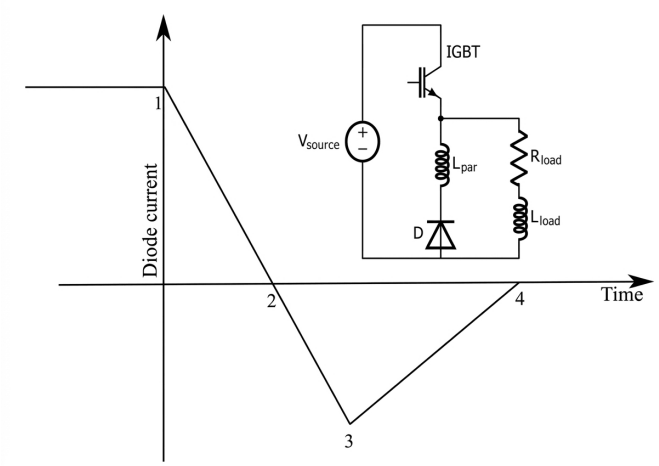


Fig. Q1.20.1

Q1.22.1 MCQ 2022 1M

A charger supplies 100 W at 20 V for charging the battery of a laptop. The power devices used in the converter inside the charger operate at a switching frequency of 200 kHz. Which power device is best suited for this purpose?

A. IGBT **B.** Thyristor **C.** MOSFET **D.** BJT

Q1.22.2 MCQ 2022 1M

For an ideal MOSFET biased in saturation, the magnitude of the small signal current gain for a common drain amplifier is

A. 0 **B.** 1 **C.** 100 **D.** infinite

Q1.20.1 NAT 2020 1M

A double pulse measurement for an inductively loaded circuit controlled by the IGBT switch is carried out to evaluate the reverse recovery characteristics of the diode, D, represented approximately as a piecewise linear plot of current vs time at diode turn-off. L_{par} is a parasitic inductance due to the wiring of the circuit, and is in series with the diode. The point on the plot (indicate your choice by entering 1, 2, 3 or 4) at which the IGBT experiences the highest current stress is _____.

Q1.18.1 MCQ 2018 1M

Four power semiconductor devices are shown in the figure along with their relevant terminals. The device(s) that can carry dc current continuously in the direction shown when gated appropriately is (are) _____.

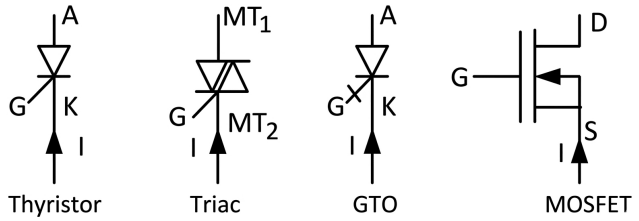


Fig. Q1.18.1

A. Triac only **B.** Triac and MOSFET
C. Triac and GTO **D.** Thyristor and Triac

Q1.17.1 MCQ 2017 1M

For the power semiconductor devices IGBT, MOSFET, Diode, and Thyristor, which one of the following statements is TRUE?

A. All the four are majority carrier devices.

B. All the four are minority carrier devices.

C. IGBT and MOSFET are majority carrier devices, whereas Diode and Thyristor are minority carrier devices.

D. MOSFET is a majority carrier device, whereas IGBT, Diode, and Thyristor are minority carrier devices.

Q1.16.1 NAT 2016 1M

A steady DC current of 100 A is flowing through a power module (S, D) as shown in Figure (a). The V–I characteristics of the IGBT (S) and the diode (D) are shown in Figures (b) and (c), respectively. The conduction power loss in the power module (S, D), in watts, is _____.

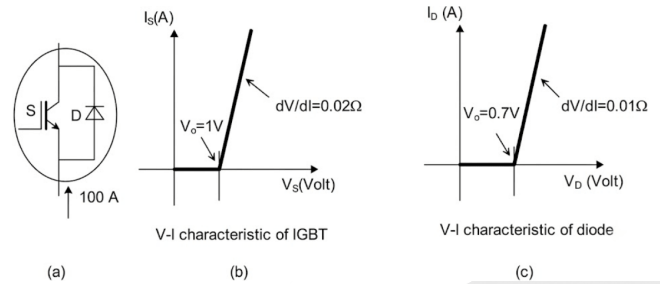


Fig. Q1.16.1

Q1.16.2 NAT 2016 1M

The voltage (v_s) across and the current (i_s) through a semiconductor switch during a turn-ON transition are shown in the figure. The energy dissipated during the turn-ON transition, in mJ, is _____.

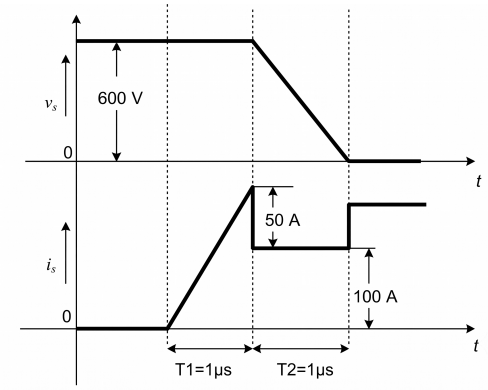


Fig. Q1.16.2

Q1.14.1 MCQ 2014 1M

Figure shows four electronic switches (i), (ii), (iii), and (iv). Which of the switches can block voltages of either polarity (applied between terminals 'a' and 'b') when the active device is in the OFF state?

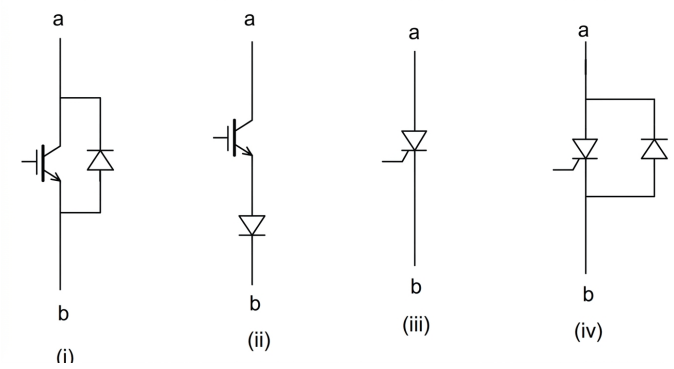


Fig. Q1.14.1

A. (i), (ii) and (iii) **B.** (ii), (iii) and (iv)
C. (ii) and (iii) **D.** (i) and (iv)

Q1.14.2 NAT 2014 2M

The SCR in the circuit shown has a latching current of 40 mA. A gate pulse of 50 μs is applied to the SCR. The maximum value of R in Ω to ensure successful firing of the SCR is _____.

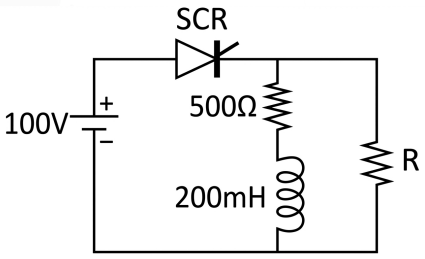


Fig. Q1.14.2

Q1.14.3 MCQ 2014 2M

Thyristor T in the figure below is initially OFF and is triggered with a single pulse of width 10 μs. It is given that

$$L = \left(\frac{100}{\pi}\right)\mu\text{H} \quad \text{and} \quad C = \left(\frac{100}{\pi}\right)\mu\text{F}.$$

Assuming the latching and holding currents of the thyristor are both zero, and the initial charge on C is zero, T conducts for

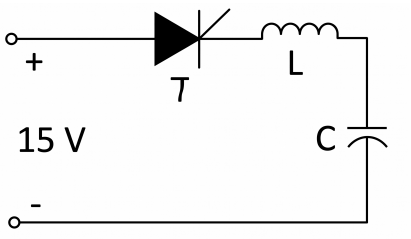


Fig. Q1.14.3

A. 10 μs **B.** 50 μs **C.** 100 μs **D.** 200 μs

Q1.12.1 MCQ 2012 1M

The typical ratio of latching current to holding current in a 20 A thyristor is

A. 5 **B.** 2 **C.** 1 **D.** 0.5

Q1.11.1 MCQ 2011 1M

Circuit turn-off time of an SCR is defined as the time

A. Taken by the SCR to turn off.

B. Required for the SCR current to become zero.

C. For which the SCR is reverse biased by the commutation circuit.

D. For which the SCR is reverse biased to reduce its current below the holding current.

Q1.10.1 MCQ 2010 1M

Figure shows a composite switch consisting of a power transistor (BJT) in series with a diode. Assuming that the transistor switch and the diode are ideal, the I–V characteristic of the composite switch is

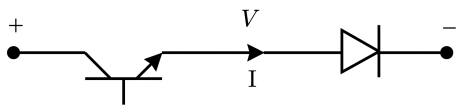
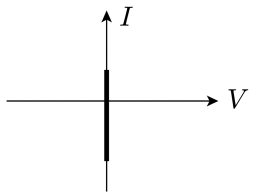
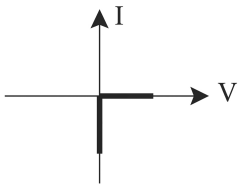
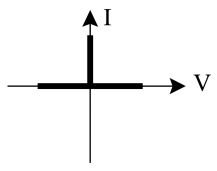
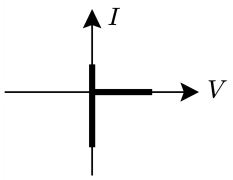


Fig. Q1.10.1

A.


B.


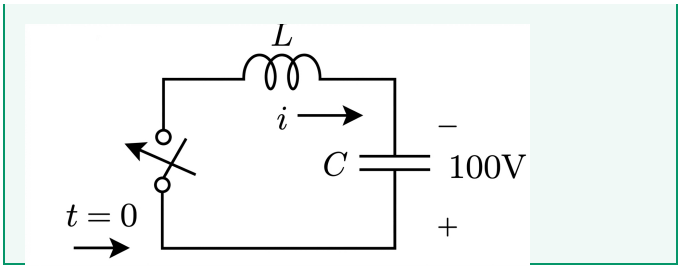
C.


D.


Common Data Questions

Common Data for Questions Q1.10.2 and Q1.10.3

The L–C circuit shown in the figure has an inductance L and a capacitance $C = 10\,\mu$F.



Q1.10.2 MCQ 2010 2M

The initial current through the inductor is zero, while the initial capacitor voltage is 100 V. The switch is closed at $t = 0$. The current i through the circuit is

A. $5\cos(5 \times 10^3 t)$ A **B.** $5\sin(10^4 t)$ A
C. $10\cos(5 \times 10^3 t)$ A **D.** $10\sin(10^4 t)$ A

Q1.10.3 MCQ 2010 2M

The L–C circuit of Q1.10.3 is used to commutate a thyristor, which is initially carrying a current of 5 A as shown in the figure below. The values and initial conditions of L and C are the same as in Q54. The switch is closed at $t = 0$. If the forward drop is negligible, the time taken for the device to turn off is

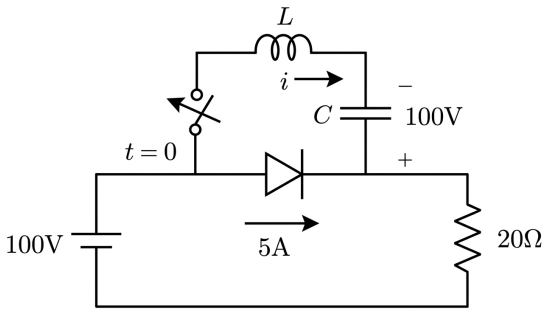


Fig. Q1.10.3

A. 52 μs **B.** 156 μs **C.** 312 μs **D.** 26 μs

Q1.09.1 MCQ 2009 2M

Match the switch arrangements on the top row to the steady-state V–I characteristics on the lower row. The steady-state operating points are shown by large black dots.

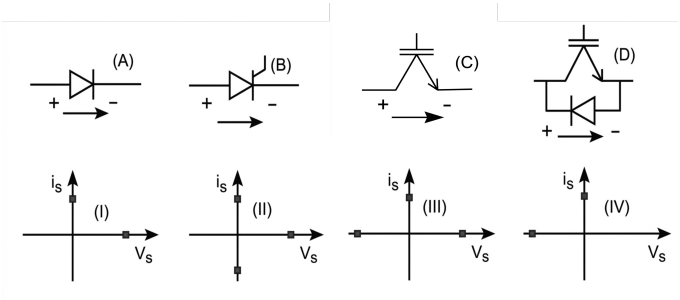


Fig. Q1.09.1

A. A–I, B–II, C–III, D–IV

B. A–II, B–IV, C–I, D–III

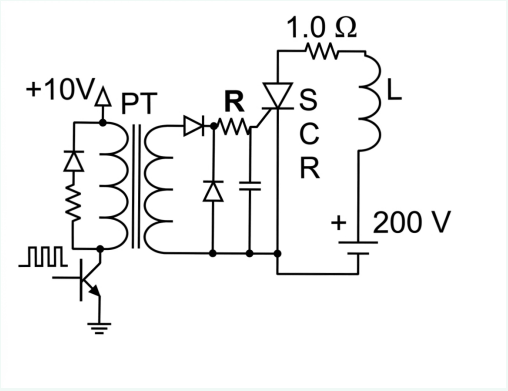
C. A–IV, B–III, C–I, D–II

D. A–IV, B–III, C–II, D–I

Common Data Questions

Common Data for Questions Q1.07.1 and Q1.07.2

A 1 : 1 pulse transformer (PT) is used to trigger the SCR in the adjacent figure. The SCR is rated at 1.5 kV, 250 A with I_L = 250 mA, I_H = 150 mA, $I_{G\,max}$ = 150 mA, and $I_{G\,min}$ = 100 mA. The SCR is connected to an inductive load, where L = 150 mH in series with a small resistance and the supply voltage is 200 V dc. The forward drops of all transistors/diodes and the gate-cathode junction during ON state are 1.0 V.



Q1.07.1 MCQ 2007 2M

The resistance R should be

A. 4.7 kΩ **B.** 470 Ω **C.** 47 Ω **D.** 4.7 Ω

Q1.07.2 MCQ 2007 2M

The minimum approximate volt-second rating of the pulse transformer suitable for triggering the SCR should be: (volt-second rating is the maximum of product of the voltage and the width of the pulse that may be applied)

A. 2000 μV-s **B.** 200 μV-s
C. 20 μV-s **D.** 2.0 μV-s

Q1.07.3 MCQ 2007 2M

The circuit in the figure is a current commutated dc-dc chopper where, Th_M is the main SCR and Th_{AUX} is the auxiliary SCR. The load current is constant at 10 A. Th_M is ON. Th_{AUX} is triggered at t = 0. Th_M is turned OFF between

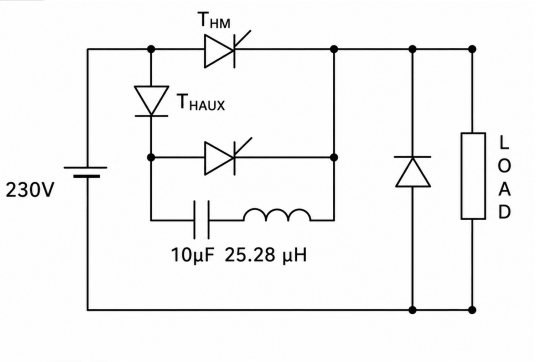


Fig. Q1.07.3

A. 0 $\mu s < t \leq 25\ \mu s$ **B.** 25 $\mu s < t \leq 50\ \mu s$
C. 50 $\mu s < t \leq 75\ \mu s$ **D.** 75 $\mu s < t \leq 100\ \mu s$

Q1.06.1 MCQ 2006 2M

An SCR having a turn-ON time of 5 μs, latching current of 50 mA, and holding current of 40 mA is triggered by a short-duration pulse and is used in the circuit shown in the figure. The minimum pulse width required to turn the SCR ON will be

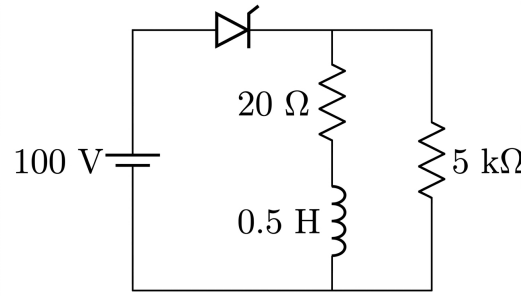


Fig. Q1.06.1

A. 251 μs **B.** 150 μs **C.** 100 μs **D.** 5 μs

Q1.06.2 MCQ 2006 2M

A voltage commutation circuit is shown in figure. If the turn off time of the SCRs is 50 μsec and a safety margin of 2 is considered, then what will be the approximate minimum value of capacitor required for proper commutation?

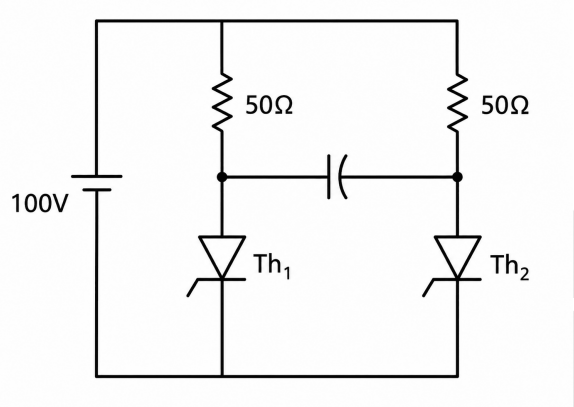


Fig. Q1.06.2

A. 2.88 μF **B.** 1.44 μF **C.** 0.9 μF **D.** 0.72 μF

Q1.05.1 MCQ 2005 1M

Assume that the N-channel MOSFET shown in the figure is ideal and that its threshold voltage is +1.0 V. The voltage V_{ab} between nodes a and b is

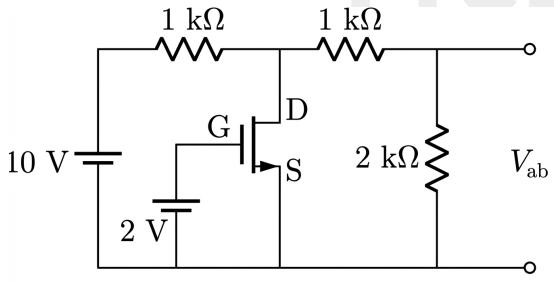


Fig. Q1.05.1

A. 5 V **B.** 2 V **C.** 1 V **D.** 0 V

Q1.05.2 MCQ 2005 1M

The conduction loss versus device current characteristic of a power MOSFET is best approximated by

A. a parabola

B. a straight line

C. a rectangular hyperbola

D. an exponentially decaying function

Q1.05.3 MCQ 2005 2M

The figure shows the voltage across a power semiconductor device and the current through the device during a switching transition. Is the transition a turn-ON transition or a turn-OFF transition? What is the energy lost during the transition?

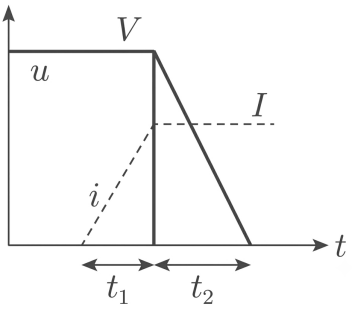


Fig. Q1.05.3

A. Turn ON, $\frac{VI}{2}(t_1 + t_2)$

B. Turn OFF, $VI(t_1 + t_2)$

C. Turn ON, $VI(t_1 + t_2)$

D. Turn OFF, $\frac{VI}{2}(t_1 + t_2)$

Q1.05.4 MCQ 2005 2M

An electronic switch S is required to block voltage of either polarity during its OFF state as shown in Figure (a). This switch is required to conduct in only one direction in its ON state as shown in Figure (b).

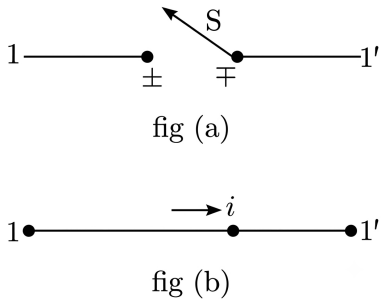


Fig. Q1.05.4

Which of the following are valid realizations of the switch S?

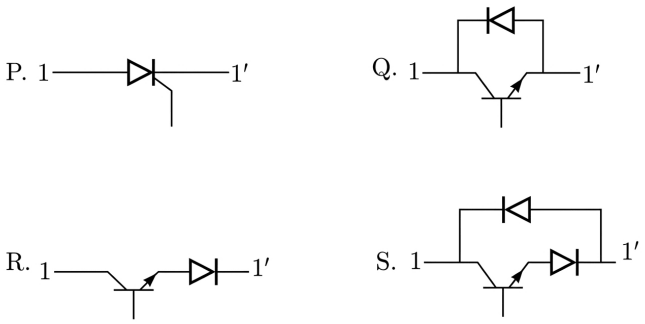


Fig. Q1.05.4

A. Only P **B.** P and Q **C.** P and R **D.** R and S

Q1.04.1 MCQ 2004 1M

The triggering circuit of a thyristor is shown in the figure. The thyristor requires a gate current of 10 mA for guaranteed turn-ON. The value of R required for the thyristor to turn ON reliably under all conditions of V_b variation is

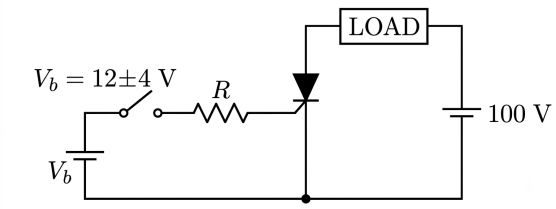


Fig. Q1.04.1

A. 1000 Ω **B.** 1600 Ω **C.** 1200 Ω **D.** 800 Ω

Q1.04.2 MCQ 2004 2M

A MOSFET rated for 15 A carries a periodic current as shown in the figure. The ON-state resistance of the MOSFET is 0.15 Ω. The average ON-state loss in the MOSFET is

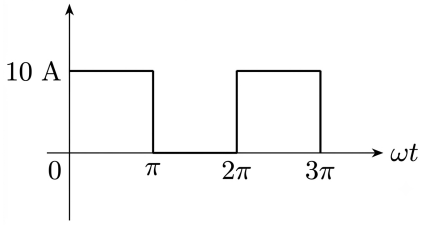


Fig. Q1.04.2

A. 33.8 W **B.** 15.0 W **C.** 7.5 W **D.** 3.8 W

Q1.03.1 MCQ 2003 1M

The variation of drain current with gate-to-source voltage (I_D–V_{GS} characteristic) of a MOSFET is shown in the figure. The MOSFET is

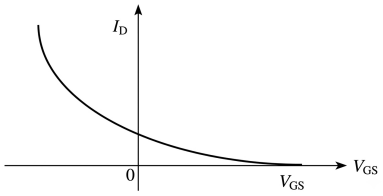


Fig. Q1.03.1

A. an n-channel depletion-mode device

B. an n-channel enhancement-mode device

C. a p-channel depletion-mode device

D. a p-channel enhancement-mode device

Q1.03.2 MCQ 2003 1M

Figure shows a MOSFET with an integral body diode. It is employed as a power switching device in the ON and OFF states through appropriate control. The ON and OFF states of the switch are given on the V_{DS}–I_S plane by

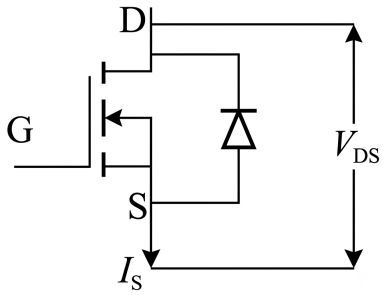
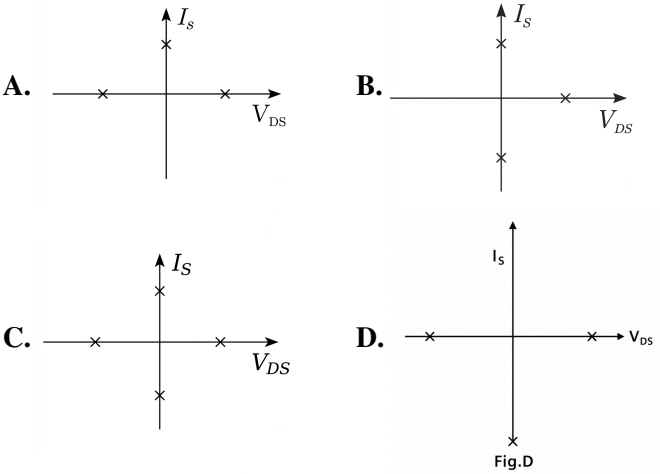


Fig. Q1.03.2



Q1.03.3 MCQ 2003 1M

Figure shows a thyristor with the standard terminations of anode (A), cathode (K), gate (G) and the different junctions named J1, J2 and J3. When the thyristor is turned on and conducting

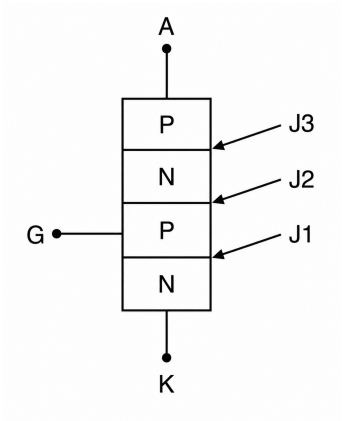


Fig. Q1.03.3

A. J_1 and J_2 are forward biased and J_3 is reverse biased

B. J_1 and J_3 are forward biased and J_2 is reverse biased

C. J_1 is forward biased and J_2 and J_3 are reverse biased

D. J_1, J_2 and J_3 are all forward biased

Q1.01.1 MCQ 2001 2M

An n-channel JFET having a pinch-off voltage (V_p) of -5 V shows a transconductance (g_m) of 1 mA/V when the applied gate-to-source voltage (V_{GS}) is -3 V. Its maximum transconductance (in mA/V) is

A. 1.5 **B.** 2 **C.** 2.5 **D.** 3

Q1.01.2 MCQ 2001 1M

The main reason for connecting a pulse transformer at the output stage of a thyristor triggering circuit is to

A. amplifying the power of the triggering pulse

B. provide electrical isolation

C. reduce the turn on time of the thyristor

D. avoid spurious triggering of the thyristor due to noise

Q1.98.1 MCQ 1998 1M

The uncontrolled electronic switch employed in power electronic converters is:

A. thyristor **B.** bipolar junction transistor
C. diode **D.** MOSFET

Q1.98.2 MCQ 1998 1M

The MOSFET switch in its on-state may be considered equivalent to:

A. resistor **B.** inductor
C. capacitor **D.** battery

Q1.98.3 MCQ 1998 1M

In a commutation circuit employed to turn off an SCR, satisfactory turn off is obtained when

A. circuit turn-off time < device turn-off time

B. circuit turn-off time > device turn-off time

C. circuit time constant > device turn-off time

D. circuit time constant < device turn-off time

Q1.97.1 NAT 1997 2M

An SCR is connected in series with a 300 V ac supply and a 300 ohms load resistor. Calculate for a firing angle of 45°,
(a) the reading of a moving coil ammeter connected in series with the load, and
(b) the reading on a moving iron voltmeter connected across the SCR.

Q1.97.2 MCQ 1997 1M

If a diode is connected in anti-parallel with a thyristor, then

A. both turn-off power loss and turn-off time decrease

B. turn-off power loss decreases but turn-off time increases

C. turn-off power loss increases, but turn-off time decreases

D. none of the above

Q1.96.1 MCQ 1996 1M

Which semiconductor power device out of the following is not a current triggered device?

A. Thyristor **B.** G.T.O.
C. Triac **D.** MOSFET

Q1.96.2 MCQ 1996 1M

The Triac can be used only in

A. inverter **B.** rectifier
C. multiquadrant chopper **D.** cycloconverter

Q1.96.3 MCQ 1996 1M

Which of the following does not cause permanent damage to an SCR?

A. High current

B. High rate of rise of current

C. High temperature rise

D. High rate of rise of voltage

Q1.95.1 NAT 1995 2M

Consider the thyristor circuit of figure. The thyristor is given a triggering pulse after every 10 ms. calculate the duration for which the thyristor remains ON after each triggering pulse. Assume ideal devices and explain briefly the basis.

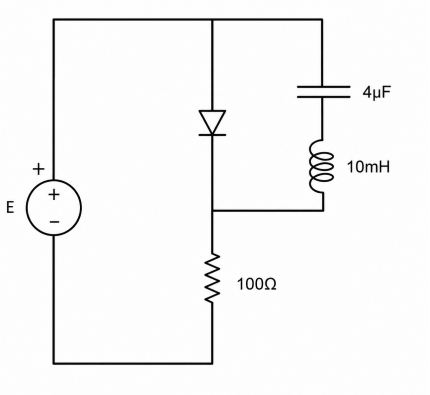


Fig. Q1.95.1

Q1.95.2 NAT 1995 1M

Figure, show two thyristors each rated 500 A (continuous) sharing a load current. Current through thyristor *y* is 120 A. The current through thyristor *x* will be nearly _____ A.

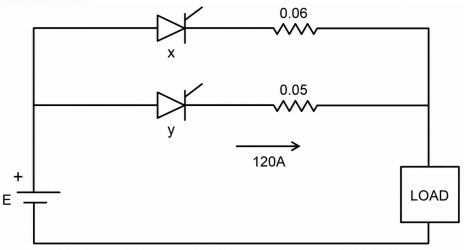


Fig. Q1.95.2

Q1.94.1 MCQ 1994 1M

A switched mode power supply operating at 20 kHz to 100 kHz range uses as the main switching element

A. Thyristor **B.** MOSFET
C. Triac **D.** UJT

Q1.94.2 NAT 1994 1M

A triac can be triggered by a gate pulse of _____ polarity.

Q1.93.1 MCQ 1993 1M

The thermal resistance between the body of a power semiconductor device and the ambient is expressed as

A. Voltage across the device divided by current through the device

B. Average power dissipated in the device divided by the square of the RMS current in the device

C. Average power dissipated in the device divided by the temperature difference from body to ambient

D. Temperature difference from body to ambient divided by average power dissipated in the device

Q1.92.1 MCQ 1992 1M

In the circuit of figure, the switch 'S' is closed at $t = 0$ with $i_L(0) = 0$ and $v_c(0) = 0$. In the steady state v_c equals

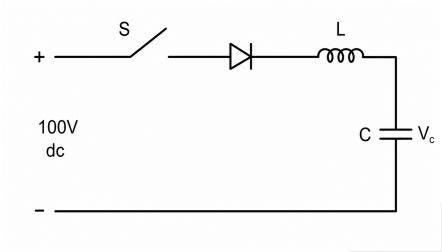


Fig. Q1.92.1

A. 200 V **B.** 100 V **C.** zero **D.** −100 V

Q1.91.1 MCQ 1991 1M

The operating state that distinguishes a silicon controlled rectifier (SCR) from a diode is

A. Forward conduction state **B.** Forward blocking state
C. Reverse conduction state **D.** Reverse blocking state

Q1.91.2 MCQ 1991 1M

Match the functions of the following protective elements in SCR applications:

SCR rating	Protective element
(A) di/dt limit	(P) Snubber
(B) dv/dt limit	(Q) Heat sink
(C) i^2t limit	(R) Series reactor
(D) Junction temperature limit	(S) To avoid runaway speeds on no-load

Answer Key

Q1.26.1 C	Q1.26.2 B	Q1.24.1 D
Q1.23.1 A,D	Q1.23.2 7.1 to 7.5	Q1.22.1 C
Q1.22.2 D	Q1.20.1 3 to 3	Q1.18.1 B
Q1.17.1 D	Q1.16.1 169 to 171	Q1.16.2 75
Q1.14.1 C	Q1.14.2 6055 to 6065	Q1.14.3 C
Q1.12.1 B	Q1.11.1 C	Q1.10.1 C
Q1.10.2 D	Q1.10.3 A	Q1.09.1 C
Q1.07.1 C	Q1.07.2 A	Q1.07.3 C
Q1.06.1 B	Q1.06.2 A	Q1.05.1 D
Q1.05.2 A	Q1.05.3 A	Q1.05.4 C
Q1.04.1 D	Q1.04.2 C	Q1.03.1 C
Q1.03.2 B	Q1.03.3 D	Q1.01.1 B
Q1.01.2 B	Q1.98.1 C	Q1.98.2 A
Q1.98.3 B	Q1.97.1 0.384,221.46	Q1.97.2 C
Q1.96.1 D	Q1.96.2 D	Q1.96.3 D
Q1.95.1 0.733ms	Q1.95.2 100	Q1.94.1 B
Q1.94.2 ±	Q1.93.1 D	Q1.92.1 A
Q1.91.1 B	Q1.91.2 R,P,S,Q	

UNIT

II

AC to DC Converters/Phase Controlled Rectifiers

Topics

1. Single-Phase Rectifier:
   (a) Half Wave Rectifier with R/RE Loads
   (b) Filters and Rectifiers with Pure Inductance(L) Load
   (c) Half Wave with RLE Loads
   (d) Half Wave Rectifiers with Free wheeling diode
   (e) Full Wave Rectifier with R/RE Loads
   (f) Full Wave Rectifier with L/RL/RLE Loads
   (g) Full Wave Rectifiers with Free wheeling diode
2. Three-Phase Rectifier:
   (a) Half Wave Rectifier with R/RL/RLE Loads
   (b) Full Wave Rectifier with RL/RLE Load with Phase Voltage
   (c) Full Wave Rectifier with RL/RLE Load with Line Voltage
   (d) Circuit Turn off time concept
   (e) Full wave Semi-Converter
   (f) Effect of Source Inductance in Rectifiers
   (g) Fourier Series Analysis of Supply Currents
3. Line-Commutated AC to DC Converters
4. Bidirectional AC to DC Voltage Source Converters
5. Line Current Harmonics, Power Factor and Distortion Factor

Unit Snapshot*

Stream: EE	**Questions:** 84	**MCQ:** 54	**MSQ:** 0	**NAT:** 30
1M: 38	**2M:** 43	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 1M(2023)–8M(2011)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 NAT 2026 1M

Consider the circuit shown. Assume that the diode D is ideal. The supply voltage is $v_s = 325\sin(2\pi \cdot 50t)$ V, with $L = 500\,\mu$H and $R = 10\,\Omega$. The peak diode current (in amperes) is _____ (Round off to one decimal place.)

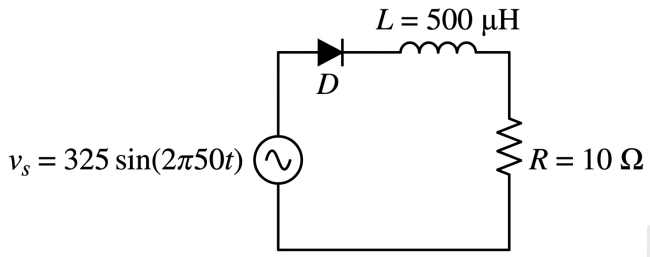


Fig. Q2.26.1

Q2.26.2 NAT 2026 2M

Consider the circuit shown. Assume that the diode (D) is ideal. Given $v_s = 100\sin(2\pi 50t)$ V, $V_{dc} = 50$ V, and $R = 10\ \Omega$, the average value of the current through the diode is _____ A (round off to two decimal places)

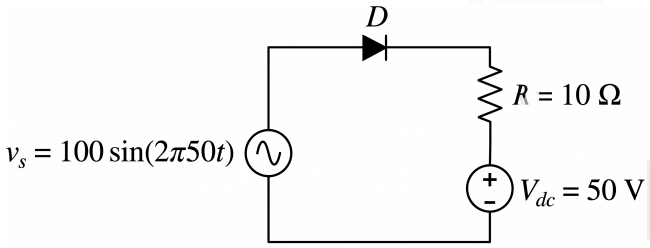
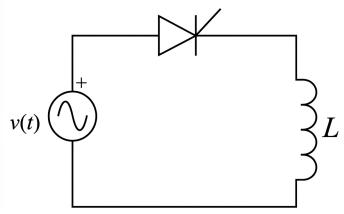


Fig. Q2.26.2

Q2.25.1 MCQ 2025 2M

An ideal sinusoidal voltage source $v(t) = 230\sqrt{2}\sin(2\pi \times 50t)$ V feeds an ideal inductor L through an ideal SCR with firing angle $\alpha = 0°$. If $L = 100$ mH, then the peak of the inductor current, in ampere, is closest to



A. 20.71 **B.** 0 **C.** 10.35 **D.** 7.32

Q2.25.2 MCQ 2025 2M

In the following circuit, the average voltage

$$V_o = 400\left(1 + \frac{\cos\alpha}{3}\right)\ \text{V},$$

where α is the firing angle. If the power dissipated in the resistor is 64 W, then the closest value of α in degrees is

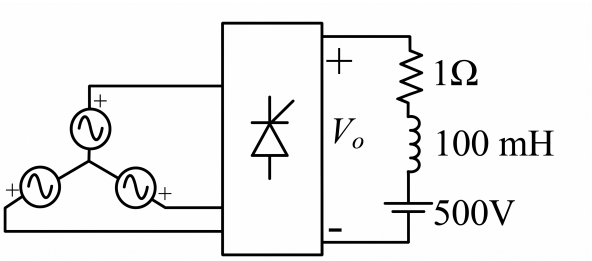


Fig. Q2.25.2

A. 35.9 **B.** 46.4 **C.** 41.4 **D.** 0

Q2.25.3 MCQ 2025 1M

The input voltage $v(t)$ and current $i(t)$ of a converter are given by

$$v(t) = 300\sin(\omega t)\ \text{V},$$

$$i(t) = 10\sin\left(\omega t - \frac{\pi}{6}\right) + 2\sin\left(3\omega t + \frac{\pi}{6}\right) + \sin\left(5\omega t + \frac{\pi}{2}\right)\ \text{A},$$

where $\omega = 2\pi \times 50$ rad/s. The input power factor of the converter is closest to

A. 0.845 **B.** 0.867 **C.** 0.887 **D.** 1.0

Q2.24.1 NAT 2024 2M

A single-phase half-controlled bridge converter supplies an inductive load with ripple-free load current. The triggering angle of the converter is 60°. The ratio of the rms value of the fundamental component of the input current to the rms value of the total input current of the bridge is _____ (rounded off to 3 decimal places).

Q2.24.2 NAT 2024 2M

A single-phase full-controlled thyristor converter bridge is used for regenerative braking of a separately excited DC motor with the following specifications:

Rated armature voltage	210 V
Rated armature current	10 A
Rated speed	1200 rpm
Armature resistance	1 Ω
Input to the converter bridge	240 V at 50 Hz
The armature of the DC motor is fed from the full-controlled bridge and the field current is kept constant.	

Assume that the motor is running at 600 rpm and the armature terminals of the motor are suitably reversed for regenerative braking. If the armature current of the motor is to be maintained at the rated value, the triggering angle of the converter bridge in degrees should be _____ (rounded off to two decimal places).

Q2.23.1 NAT 2023 1M

The single phase rectifier consisting of three thyristors T_1, T_2, T_3 and a diode D_1 feed power to a 10 A constant current load. T_1 and T_3 are fired at $\alpha = 60°$ and T_2 is fired at $\alpha = 240°$. The reference for α is the positive zero crossing of V_{in}. The average voltage V_o across the load in volts is _____ (Round off to 2 decimal places).

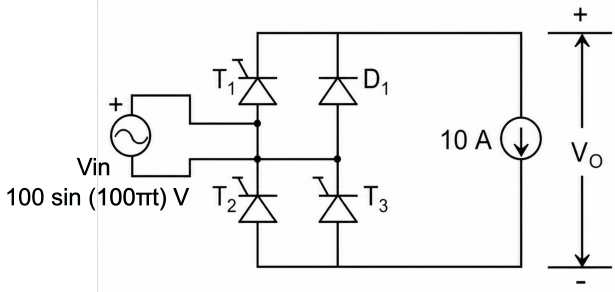


Fig. Q2.23.1

Q2.22.1 NAT 2022 1M

A single-phase full-bridge diode rectifier feeds a resistive load of 50 Ω from a 200 V, 50 Hz single-phase AC supply. If the diodes are ideal, then the active power, in watts, drawn by the load is _____ (*round off to nearest integer*).

Q2.22.2 NAT 2022 1M

The voltage at the input of an AC–DC rectifier is given by

$$v(t) = 230\sqrt{2}\sin\omega t$$

where $\omega = 2\pi \times 50$ rad/s. The input current drawn by the rectifier is

$$i(t) = 10\sin\left(\omega t - \frac{\pi}{3}\right) + 4\sin\left(3\omega t - \frac{\pi}{6}\right) + 3\sin\left(5\omega t - \frac{\pi}{3}\right)$$

The input power factor (rounded off to two decimal places) is _____ lag.

Q2.22.3 NAT 2022 2M

For the ideal AC-DC rectifier circuit shown in the figure below, the load current magnitude is

$$I_{dc} = 15 \text{ A}$$

and is ripple free. The thyristors are fired with a delay angle of 45°. The amplitude of the fundamental component of the source current, in amperes, is _____. *(round off to two decimal places)*

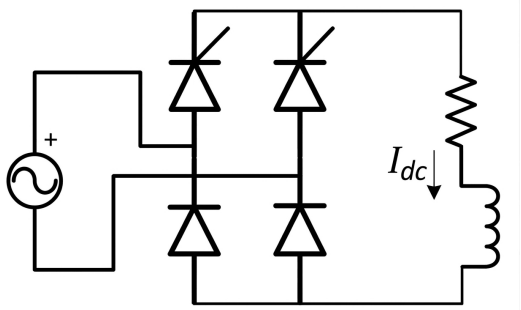


Fig. Q2.22.3

Q2.21.1 NAT 2021 2M

In the circuit shown, the input V_i is a sinusoidal AC voltage having an RMS value of 230 V ± 20%. The worst-case peak-inverse voltage seen across any diode is _____ V. (Round off to 2 decimal places.)

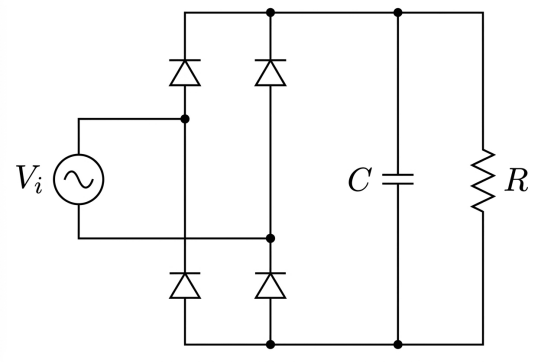


Fig. Q2.21.1

Q2.20.1 MCQ 2020 1M

A single-phase, full-bridge diode rectifier fed from a 230 V, 50 Hz sinusoidal source supplies a series combination of finite resistance, R, and a very large inductance, L. The two most dominant frequency components in the source current are:

A. 50 Hz, 0 Hz **B.** 50 Hz, 100 Hz
C. 50 Hz, 150 Hz **D.** 150 Hz, 250 Hz

Q2.20.2 NAT 2020 2M

A single-phase, full-bridge, fully controlled thyristor rectifier feeds a load comprising a 10 Ω resistance in series with a very large inductance. The rectifier is fed from an ideal 230 V, 50 Hz sinusoidal source through cables which have negligible internal resistance and a total inductance of 2.28 mH. If the thyristors are triggered at an angle $\alpha = 45°$, the commutation overlap angle in degree (rounded off to 2 decimal places) is _____.

Q2.19.1 MCQ 2019 1M

A six-pulse thyristor bridge rectifier is connected to a balanced three-phase, 50 Hz AC source. Assuming that the DC output current of the rectifier is constant, the lowest harmonic component in the AC input current is _____.

A. 100 Hz **B.** 150 Hz **C.** 250 Hz **D.** 300 Hz

Q2.19.2 NAT 2019 2M

A fully-controlled three-phase bridge converter is working from a 415 V, 50 Hz AC supply. It is supplying constant current of 100 A at 400 V to a DC load. Assume large inductive smoothing and neglect overlap. The rms value of the AC line current in amperes (round off to two decimal places) is _____.

Q2.19.3 NAT 2019 2M

A single-phase fully-controlled thyristor converter is used to obtain an average voltage of 180 V with 10 A constant current to feed a DC load. It is fed from single-phase AC supply of 230 V, 50 Hz. Neglect the source impedance. The power factor (round off to two decimal places) of AC mains is _____.

Q2.18.1 MCQ 2018 1M

A single phase fully controlled rectifier is supplying a load with an anti-parallel diode as shown in the figure. All switches and diodes are ideal. Which one of the following is true for instantaneous load voltage and current?

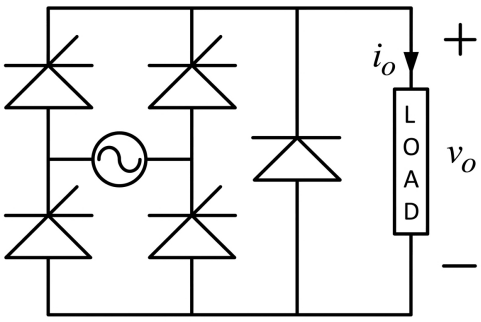


Fig. Q2.18.1

A. $v_o \geq 0$ & $i_o < 0$ **B.** $v_o < 0$ & $i_o < 0$
C. $v_o \geq 0$ & $i_o \geq 0$ **D.** $v_o < 0$ & $i_o \geq 0$

Q2.18.2 NAT 2018 1M

The waveform of the current drawn by a semi-converter from a sinusoidal AC voltage source is shown in the figure. If I_0 = 20 A, the rms value of fundamental component of the current is _____ A (up to 2 decimal places).

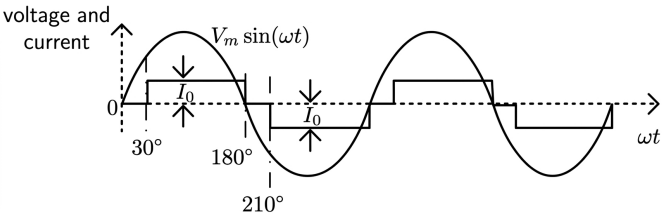


Fig. Q2.18.2

Q2.18.3 NAT 2018 2M

A phase controlled single phase rectifier, supplied by an AC source, feeds power to an R-L-E load as shown in the figure. The rectifier output voltage has an average value given by

$$V_o = \frac{V_m}{2\pi}(3 + \cos\alpha),$$

where $V_m = 80\pi$ volts and α is the firing angle. If the power delivered to the lossless battery is 1600 W, α in degree is _____ (up to 2 decimal places).

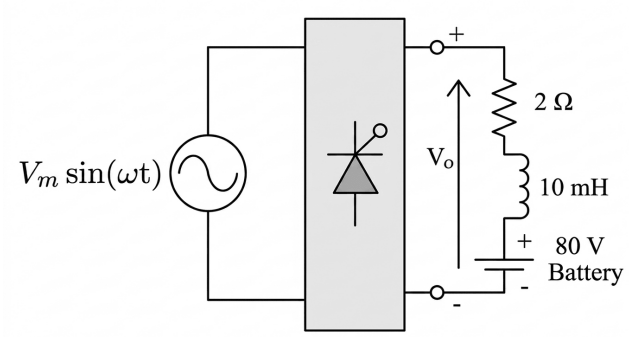


Fig. Q2.18.3

Q2.17.1 NAT 2017 2M

The figure below shows an uncontrolled diode bridge rectifier supplied from a 220V, 50 Hz, 1-phase AC source. The load draws a constant current $I_o = 14$ A. The conduction angle of the diode D_1 in degrees (rounded off to two decimal places) is _____.

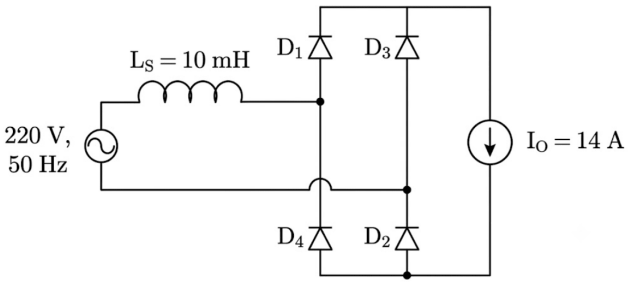


Fig. Q2.17.1

Q2.17.2 NAT 2017 1M

The figure below shows the circuit diagram of a controlled rectifier supplied from a 230 V, 50 Hz, 1-phase voltage source and a 10 : 1 ideal transformer. Assume that all devices are ideal. The firing angles of the thyristors T_1 and T_2 are 90° and 270°, respectively.

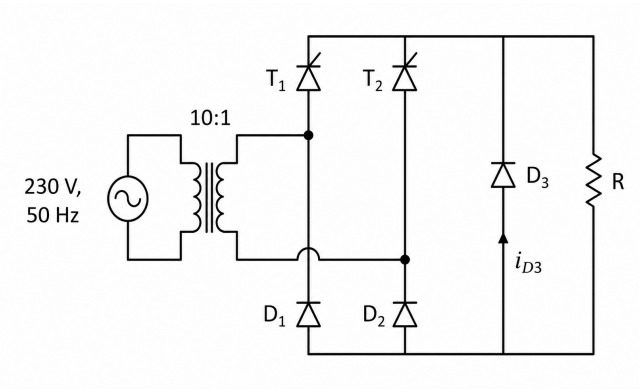


Fig. Q2.17.2

The RMS value of the current through diode D_3 in amperes is _____.

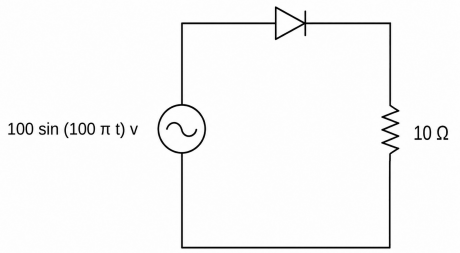
Q2.17.3 MCQ 2017 1M

A phase-controlled, single-phase, full-bridge converter is supplying a highly inductive DC load. The converter is fed from a 230 V, 50 Hz AC source. The fundamental frequency in Hz of the voltage ripple on the DC side is

A. 25 **B.** 50 **C.** 100 **D.** 300

Q2.17.4 NAT 2017 2M

In the circuit shown in the figure, the diode used is ideal. The input power factor is _____ (Give the answer up to two decimal places).



Q2.16.1 MCQ 2016 2M

A single-phase thyristor-bridge rectifier is fed from a 230 V, 50 Hz, single-phase AC mains. If it is delivering a constant DC current of 10 A, at firing angle of 30°, then value of the power factor at AC mains is

A. 0.87 **B.** 0.9 **C.** 0.78 **D.** 0.45

Q2.16.2 NAT 2016 2M

A single-phase bi-directional voltage source converter (VSC) is shown in the figure below. All devices are ideal. It is used to charge a battery at 400 V with power of 5 kW from a source V_s = 220 V (rms), 50 Hz sinusoidal AC mains at unity p.f. If its AC side interfacing inductor is 5 mH and the switches are operated at 20 kHz, then the phase shift (δ) between AC mains voltage (V_s) and fundamental AC rms VSC voltage (V_{C1}), in degree, is _____.

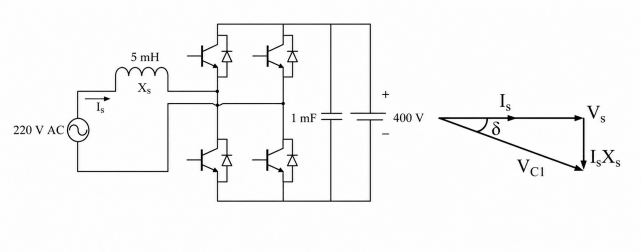


Fig. Q2.16.2

Q2.16.3 NAT 2016 2M

A full-bridge converter supplying an RLE load is shown in figure. The firing angle of the bridge converter is 120°. The supply voltage $v_m(t) = 200\pi \sin(100\pi t)$ V, R = 20 Ω, E = 800 V. The inductor L is large enough to make the output current I_L smooth dc current. Switches are lossless. The real power fed back to the source, in kW, is _____.

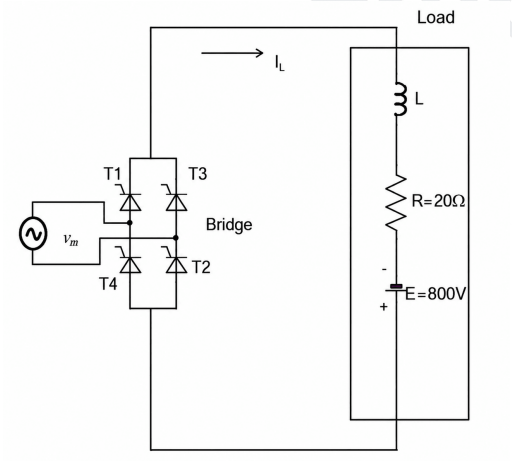


Fig. Q2.16.3

Q2.16.4 NAT 2016 1M

A three-phase diode bridge rectifier is feeding a constant DC current of 100 A to a highly inductive load. If three-phase, 415 V, 50 Hz AC source is supplying to this bridge rectifier then the rms value of the current in each diode, in ampere, is _____.

Q2.15.1 MCQ 2015 1M

Consider a HVDC link which uses thyristor based line-commutated converters as shown in the figure. For a power flow of 750 MW from System 1 to System 2, the voltages at the two ends, and the current, are given by: V_1 = 500 kV, V_2 = 485 kV and I = 1.5 kA. If the direction of power flow is to be reversed (that is, from System 2 to System 1) without changing the electrical connections, then which one of the following combinations is feasible?

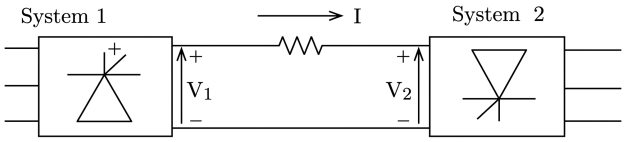


Fig. Q2.15.1

A. $V_1 = -500$ kV, $V_2 = -485$ kV and $I = 1.5$ kA

B. $V_1 = -485$ kV, $V_2 = -500$ kV and $I = 1.5$ kA

C. $V_1 = 500$ kV, $V_2 = 485$ kV and $I = -1.5$ kA

D. $V_1 = -500$ kV, $V_2 = -485$ kV and $I = -1.5$ kA

Q2.15.2 MCQ 2015 1M

In the following circuit, the input voltage V_{in} is $100\sin(100\pi t)$. For $100\pi RC = 50$, the average voltage across R (in Volts) under steady-state is nearest to _____.

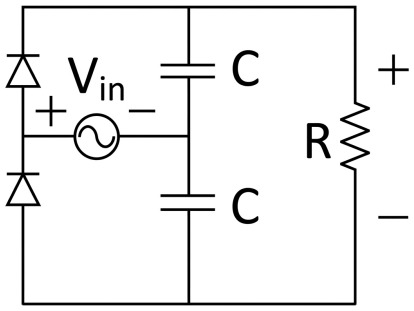


Fig. Q2.15.2

A. 100 **B.** 31.8 **C.** 200 **D.** 63.6

Q2.15.3 NAT 2015 2M

In the given rectifier, the delay angle of the thyristor T_1 measured from the positive-going zero crossing of V_s is 30°. If the input voltage V_s is $100\sin(100\pi t)$ V, the average voltage across R (in Volt) under steady-state is _____.

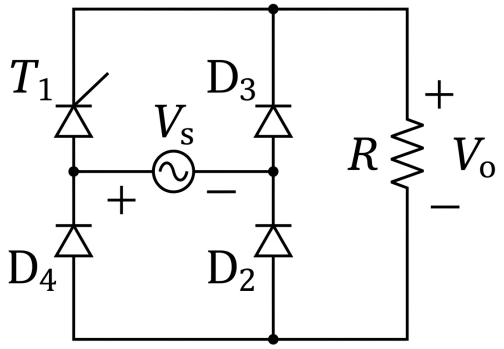


Fig. Q2.15.3

Q2.14.1 NAT 2014 1M

The figure shows the circuit of a rectifier fed from a 230 V (rms), 50 Hz sinusoidal voltage source. If we want to replace the current source with a resistor so that the rms value of the current supplied by the voltage source remains unchanged, the value of the resistance (in ohms) is _____. (Assume diodes to be ideal.)

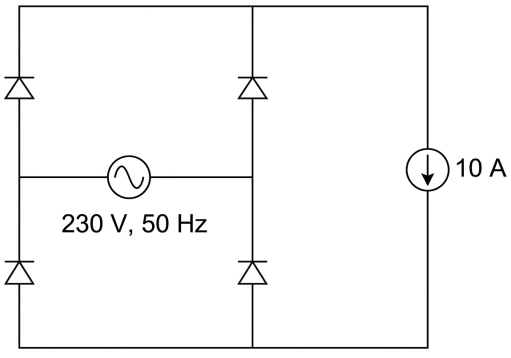


Fig. Q2.14.1

Q2.14.2 NAT 2014 2M

The figure shows the circuit diagram of a rectifier. The load consists of a resistance 10 Ω and an inductance 0.05 H connected in series. Assuming ideal thyristor and ideal diode, the thyristor firing angle (in degree) needed to obtain an average load voltage of 70V is _____

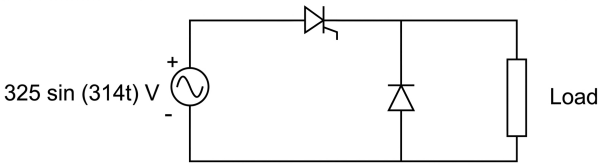


Fig. Q2.14.2

Q2.14.3 NAT 2014 2M

A fully controlled converter bridge feeds a highly inductive load with ripple free load current. The input supply (v_s) to the bridge is a sinusoidal source. Triggering angle of the bridge converter is $\alpha = 30°$. The input power factor of the bridge is _____.

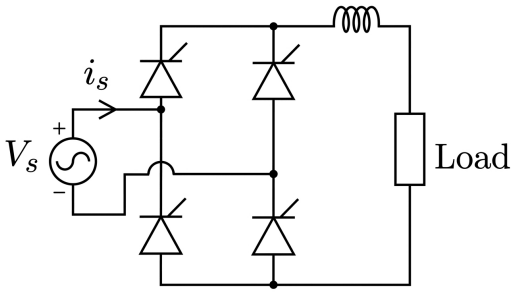


Fig. Q2.14.3

Q2.14.4 NAT 2014 2M

A diode circuit feeds an ideal inductor as shown in the figure. Given $v_s = 100\sin(\omega t)$V, where $\omega = 100\pi$ rad/s, and L = 31.83 mH. The initial value of inductor current is zero. Switch S is closed at t = 2.5 ms. The peak value of inductor current i_L (in A) in the first cycle is _____.

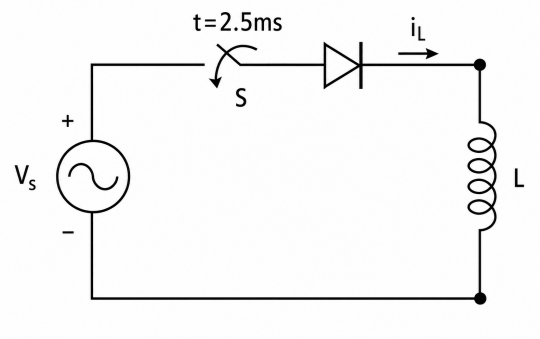


Fig. Q2.14.4

Q2.14.5 MCQ 2014 2M

A three-phase fully controlled bridge converter is fed through a star–delta transformer as shown in the figure.

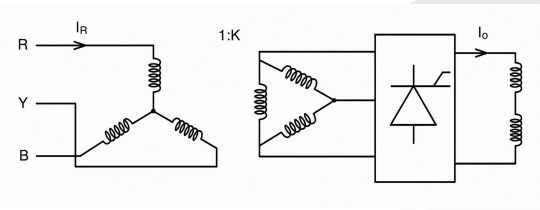
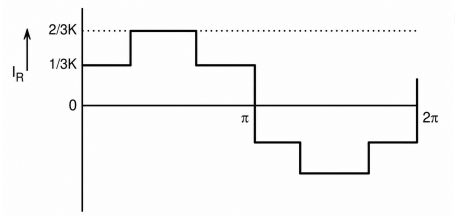


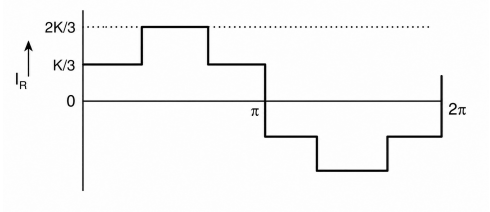
Fig. Q2.14.5

The converter is operated at a firing angle of 30°. Assuming the load current (I_o) to be virtually constant at 1 p.u. and the transformer to be an ideal one, the input phase current waveform is

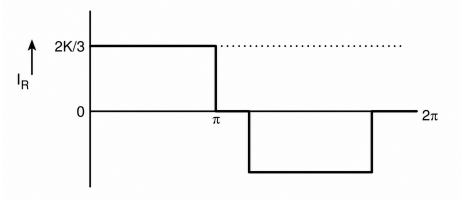
A.



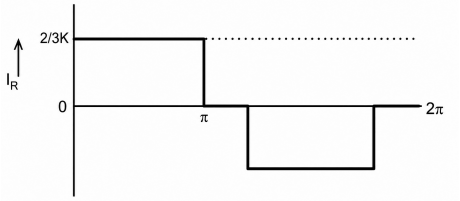
B.



C.



D.



Q2.13.1 MCQ 2013 2M

Thyristor T in the figure below is initially off and is triggered with a single pulse of width 10 μs. It is given that

$$L = \left(\frac{100}{\pi}\right) \mu\text{H}$$

and

$$C = \left(\frac{100}{\pi}\right) \mu\text{F}.$$

Assuming latching and holding currents of the thyristor are both zero and the initial charge on C is zero, T conducts for

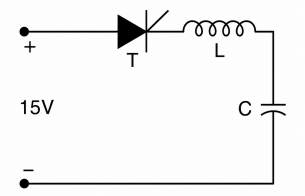


Fig. Q2.13.1

A. 10 μs **B.** 50 μs **C.** 100 μs **D.** 200 μs

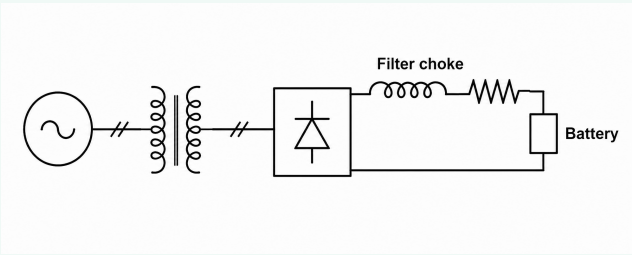
Q2.12.1 MCQ 2012 1M

A half-controlled single-phase bridge rectifier is supplying an R-L load. It is operated at a firing angle α and the load current is continuous. The fraction of cycle that the freewheeling diode conducts is

A. $\frac{1}{2}$ **B.** $\left(1 - \frac{\alpha}{\pi}\right)$ **C.** $\frac{\alpha}{2\pi}$ **D.** $\frac{\alpha}{\pi}$

Common Data Questions

Common Data for Questions Q2.11.1 and Q2.11.2
A solar energy installation utilize a three-phase bridge converter to feed energy into power system through a transformer of 400V/400 V, as shown below.



The energy is collected in a bank of 400 V battery and is connected to converter through a large filter choke of resistance 10 Ω.

Q2.11.1 MCQ 2011 2M

The maximum current through the battery will be

A. 14 A **B.** 40 A **C.** 80 A **D.** 94 A

Q2.11.2 MCQ 2011 2M

The kVA rating of the input transformer is

A. 53.2 kVA **B.** 46.0 kVA **C.** 22.6 kVA **D.** 19.6 kVA

Common Data Questions

Common Data for Questions Q2.11.3 and Q2.11.4

The input voltage given to a converter is

$$v_i = 100\sqrt{2}\sin(100\pi t)\ \text{V}$$

The current drawn by the converter is

$$i_i = 10\sqrt{2}\sin\left(100\pi t - \frac{\pi}{3}\right) + 5\sqrt{2}\sin\left(300\pi t + \frac{\pi}{4}\right) + 2\sqrt{2}\sin\left(500\pi t - \frac{\pi}{6}\right)\ \text{A}$$

Q2.11.3 MCQ 2011 2M

The input power factor of the converter is

A. 0.31 **B.** 0.44 **C.** 0.5 **D.** 0.71

Q2.11.4 MCQ 2011 2M

The active power drawn by the converter is

A. 181 W **B.** 500 W **C.** 707 W **D.** 887 W

Q2.10.1 MCQ 2010 1M

The fully controlled thyristor converter in the figure is fed from a single-phase source. When the firing angle is 0°, the dc output voltage of the converter is 300 V. What will be the output voltage for a firing angle of 60°, assuming continuous conduction?

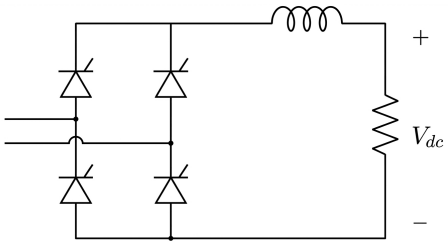


Fig. Q2.10.1

A. 150 V **B.** 210 V **C.** 300 V **D.** 100π V

Q2.09.1 MCQ 2009 1M

The circuit shows an ideal diode connected to a pure inductor and is connected to a purely sinusoidal 50Hz voltage source. Under ideal conditions the current waveform through the inductor will look like

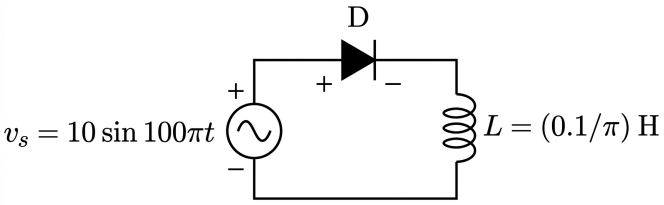
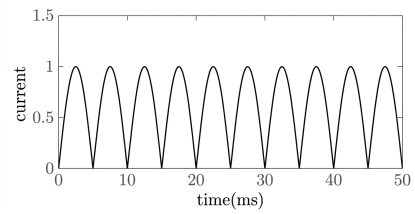
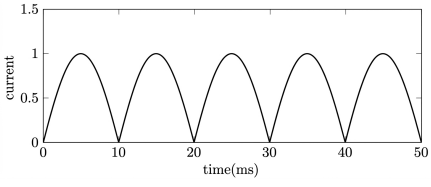


Fig. Q2.09.1

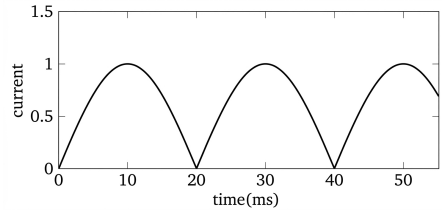
A.



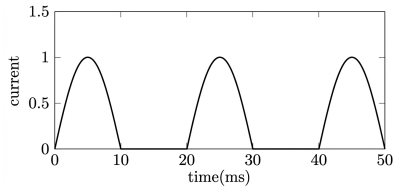
B.



C.



D.



Q2.08.1 MCQ 2008 2M

A single-phase half controlled converter shown in the figure is feeding power to highly inductive load. The converter is operating at a firing angle of 60°.

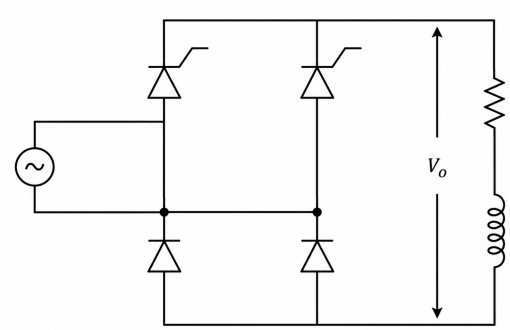
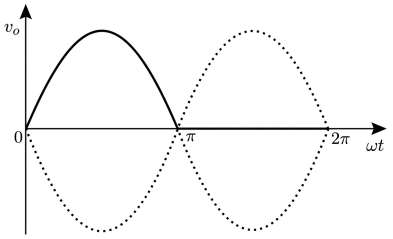


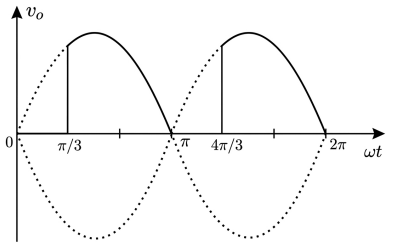
Fig. Q2.08.1

If the firing pulses are suddenly removed, the steady-state voltage (v_o) waveform of the converter will become:

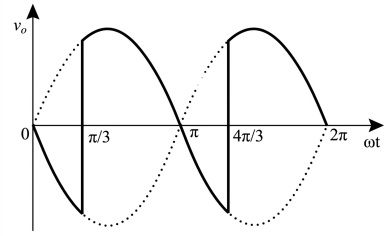
A.



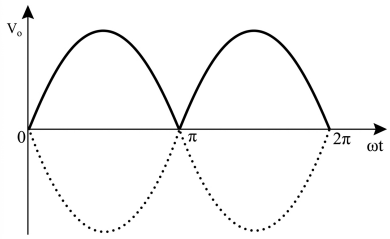
B.



C.



D.



Q2.08.2 MCQ 2008 2M

A single phase fully controlled converter bridge is used for electrical braking of a separately excited dc motor. The dc motor load is represented by an equivalent circuit as shown in the figure.

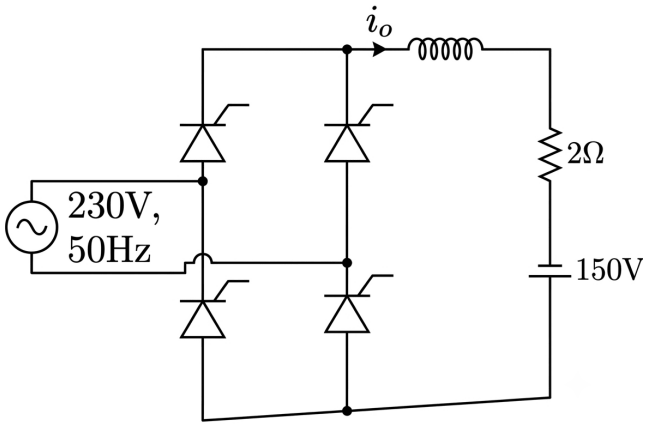


Fig. Q2.08.2

Assume that the load inductance is sufficient to ensure continuous and ripple free load current. The firing angle of the bridge for a load current of I_0 = 10 A will be

A. 44° **B.** 51° **C.** 129° **D.** 136°

Q2.08.3 MCQ 2008 2M

A three phase fully controlled bridge converter is feeding a load drawing a constant and ripple free load current of 10 A at a firing angle of 30°. The approximate Total Harmonic Distortion (%THD) and the rms value of fun-

damental component of the input current will respectively be

A. 31% and 6.8 A **B.** 31% and 7.8 A
C. 66% and 6.8 A **D.** 66% and 7.8 A

Q2.08.4 MCQ 2008 2M

A single phase fully controlled bridge converter supplies a load drawing constant and ripple free load current. If the triggering angle is 30°, the input power factor will be

A. 0.65 **B.** 0.78 **C.** 0.85 **D.** 0.866

Q2.07.1 MCQ 2007 1M

A single-phase fully controlled thyristor bridge ac-dc converter is operating at a firing angle of 25° and an overlap angle of 10° with constant dc output current of 20 A. The fundamental power factor (displacement factor) at input ac mains is

A. 0.78 **B.** 0.827 **C.** 0.866 **D.** 0.9

Q2.07.2 MCQ 2007 1M

A three-phase, fully controlled thyristor bridge converter is used as a line-commutated inverter to feed 50 kW power at 420 V dc to a three-phase, 415 V (line), 50 Hz ac mains. Consider the dc-link current to be constant. The rms current of the thyristor is

A. 119.05 A **B.** 79.37 A **C.** 68.73 A **D.** 39.68 A

Q2.07.3 MCQ 2007 1M

A single phase full-wave half-controlled bridge converter feeds an inductive load. The two SCRs in the converter are connected to a common DC bus. The converter has to have a freewheeling diode.

A. because the converter inherently does not provide for free-wheeling.

B. because the converter does not provide for free-wheeling for high values of triggering angles.

C. or else the free-wheeling action of the converter will cause shorting of the AC supply.

D. or else if a gate pulse to one of the SCRs is missed, it will subsequently cause a high load current in the other SCR.

Q2.07.4 MCQ 2007 2M

A three-phase, 440 V, 50 Hz ac mains fed thyristor bridge is feeding a 440 V dc, 15 kW, 1500 rpm separately excited dc motor with a ripple free continuous current in dc link under all operating conditions. Neglecting the losses, the power factor of the ac mains at half the rated speed, is

A. 0.354 **B.** 0.372 **C.** 0.90 **D.** 0.955

Q2.07.5 MCQ 2007 2M

A single-phase, 230 V, 50 Hz ac mains fed step down transformer (4 : 1) is supplying power to a half-wave uncontrolled ac-dc converter used for charging a battery (12 V dc) with the series current limiting resistor being 19.04 Ω. The charging current is

A. 2.43 A **B.** 1.65 A **C.** 1.22 A **D.** 1.0 A

Q2.07.6 MCQ 2007 2M

In the circuit of adjacent figure the diode connects the ac source to a pure inductance L. The diode conducts for

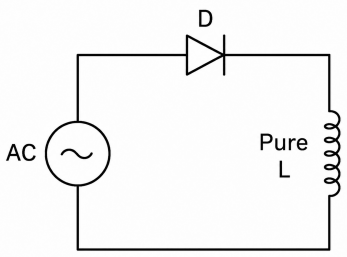


Fig. Q2.07.6

A. 90° **B.** 180° **C.** 270° **D.** 360°

Q2.06.1 MCQ 2006 1M

A single-phase half wave uncontrolled converter circuit is shown in figure. A 2-winding transformer is used at the input for isolation. Assuming the load current to be constant and $V = V_m \sin \omega t$, the current waveform through diode D_2 will be

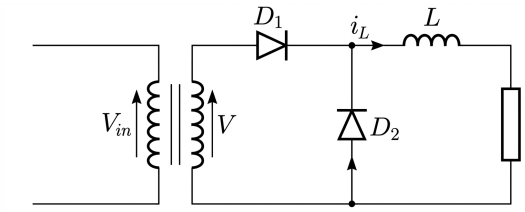
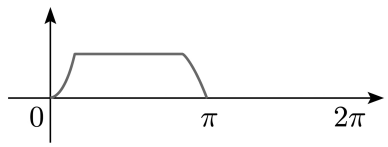
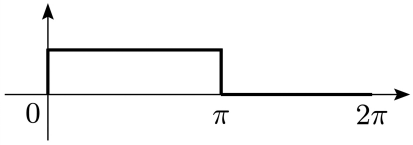


Fig. Q2.06.1

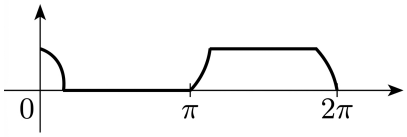
A.



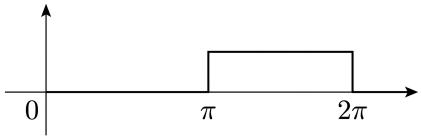
B.



C.



D.



Q2.06.2 MCQ 2006 2M

A 3-phase fully controlled bridge converter with free wheeling diode is fed from 400 V, 50 Hz AC source and is operating at a firing angle of 60°. The load current is assumed constant at 10 A due to high load inductance. The input displacement factor (IDF) and the input power factor (IPF) of the converter will be

A. IDF = 0.867; IPF = 0.828

B. IDF = 0.867; IPF = 0.552

C. IDF = 0.5; IPF = 0.478

D. IDF = 0.5; IPF = 0.318

Q2.06.3 MCQ 2006 2M

A solar cell of 350V is feeding power to an ac supply of 440V, 50 Hz through a 3-phase fully controlled bridge converter. A large inductance is connected in the dc circuit to maintain the dc current at 20 A. If the solar cell resistance is 0.5 Ω, then each thyristor will be reverse biased for a period of

A. 55° **B.** 120° **C.** 60° **D.** 125°

Q2.06.4 MCQ 2006 2M

A single-phase bridge converter is used to charge a battery of 200 V having an internal resistance of 0.2 Ω as shown in figure. The SCRs are triggered by a constant dc signal. If SCR_2 gets open circuited, what will be the average charging current?

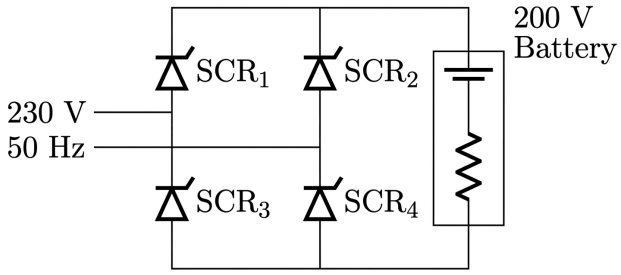


Fig. Q2.06.4

A. 23.8 A **B.** 15 A **C.** 11.9 A **D.** 3.54 A

Q2.05.1 MCQ 2005 1M

A three-phase diode bridge rectifier is fed from a 400 V RMS, 50 Hz, three-phase AC source. If the load is purely resistive, then peak instantaneous output voltage is equal to

A. 400 V **B.** $400\sqrt{2}$ V **C.** $400\sqrt{\frac{2}{3}}$ V **D.** $\frac{400}{\sqrt{3}}$ V

Q2.05.2 MCQ 2005 2M

Consider a phase-controlled converter shown in the figure. The thyristor is fired at an angle α in every positive half cycle of the input voltage. If the peak value of the instantaneous output voltage equals 230 V, the firing angle α is close to

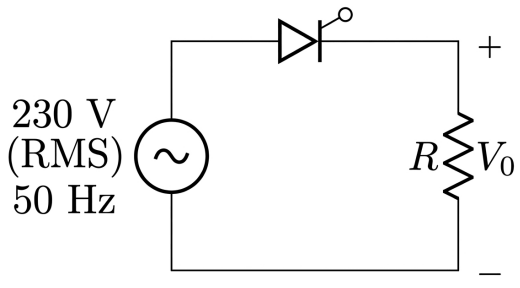


Fig. Q2.05.2

A. 45° **B.** 135° **C.** 90° **D.** 83.6°

Q2.04.1 MCQ 2004 1M

The circuit in figure shows a 3-phase half-wave rectifier. The source is a symmetrical, 3-phase four-wire system. The line-to-line voltage of the source is 100 V. The supply frequency is 400 Hz. The ripple frequency at the output is

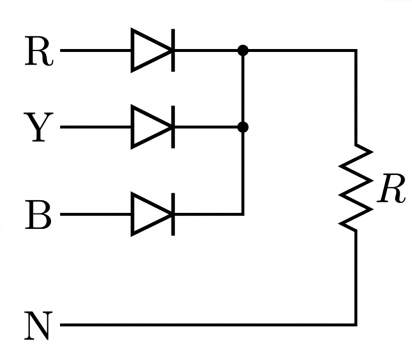


Fig. Q2.04.1

A. 400 Hz **B.** 800 Hz **C.** 1200 Hz **D.** 2400 Hz

Q2.03.1 MCQ 2003 1M

A fully controlled natural commutated 3-phase bridge rectifier is operating with a firing angle $\alpha = 30°$, The peak to peak voltage ripple expressed as a ratio of the peak output dc voltage at the output of the converter bridge is

A. 0.5 **B.** $\frac{\sqrt{3}}{2}$ **C.** $\left(1 - \frac{\sqrt{3}}{2}\right)$ **D.** $\sqrt{3} - 1$

Q2.03.2 MCQ 2003 2M

A single-phase fully controlled bridge converter feeding a constant DC current load I_{dc} is shown in the figure.The control angle $\alpha = 30°$. The output dc voltage wave shape will be as shown in

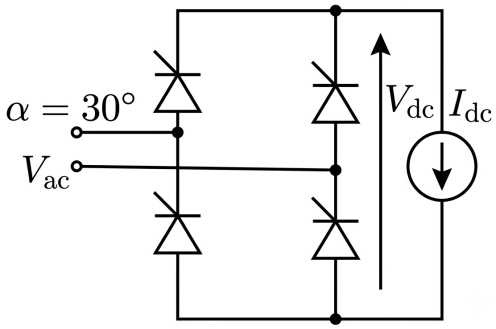


Fig. Q2.03.2

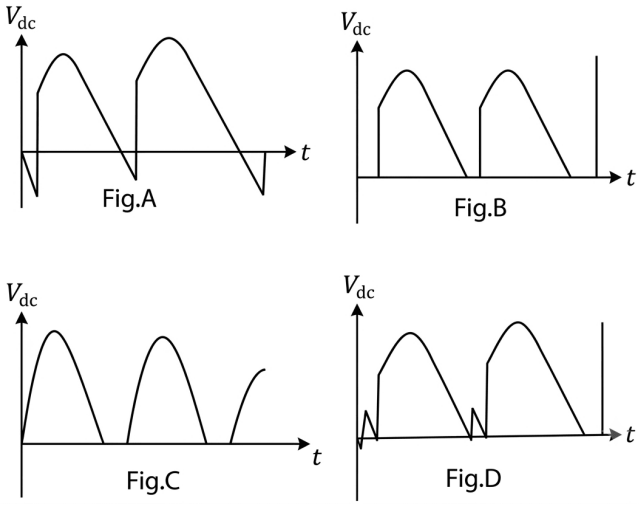


Fig. Q2.03.2

A. Fig. A **B.** Fig. B **C.** Fig. C **D.** Fig. D

Q2.02.1 MCQ 2002 1M

A six pulse thyristor rectifier bridge is connected to a balanced 50 Hz three phase ac source. Assuming that the dc output current of the rectifier is constant, the lowest frequency harmonic component in the ac source line current is

A. 100 Hz **B.** 150 Hz **C.** 250 Hz **D.** 300 Hz

Q2.02.2 MCQ 2002 2M

A three phase thyristor bridge rectifier is used in a HVDC link. The firing angle α (as measured from the point of natural commutation) is constrained to lie between 5° and 30°. If the dc side current and ac side voltage magnitudes are constant, which of the following statements is true (neglect harmonics in the ac side currents and commutation overlap in your analysis)

A. Reactive power absorbed by the rectifier is maximum when $\alpha = 5°$.

B. Reactive power absorbed by the rectifier is maximum when $\alpha = 30°$.

C. Reactive power absorbed by the rectifier is maximum when $\alpha = 15°$.

D. Reactive power absorbed by the rectifier is maximum when $\alpha = 10°$.

Q2.02.3 MCQ 2002 2M

In the single phase diode bridge rectifier shown in figure 2.02.4, the load resistor is $R = 50\,\Omega$. The source voltage is $v = 200\sin(\omega t)$, where $\omega = 2\pi \times 50$ radians per second. The power dissipated in the load resistor R is

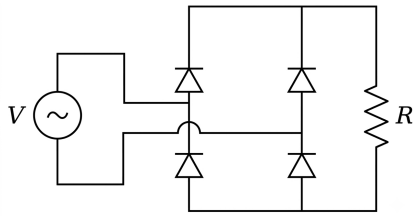


Fig. Q2.02.3

A. $\frac{3200}{\pi}$W **B.** $\frac{400}{\pi}$W **C.** 400 W **D.** 800 W

Q2.01.1 MCQ 2001 2M

A half-wave thyristor converter supplies a purely inductive load, as shown in Fig. 2.01.1. If the triggering angle of the thyristor is 120°, the extinction angle will be

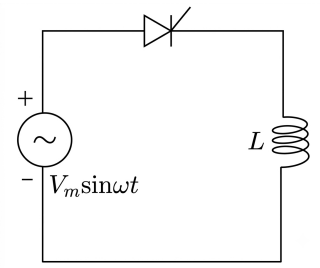


Fig. Q2.01.1

A. 240° **B.** 180° **C.** 200° **D.** 120°

Q2.01.2 MCQ 2001 1M

AC-to-DC circulating current dual converters is operated with the following relationship between their triggering angles (α_1 and α_2).

A. $\alpha_1 + \alpha_2 = 180°$ **B.** $\alpha_1 + \alpha_2 = 360°$
C. $\alpha_1 - \alpha_2 = 180°$ **D.** $\alpha_1 + \alpha_2 = 90°$

Q2.00.1 MCQ 2000 1M

A three phase semi-converter feeds the armature of a separately excited dc motor, supplying a non-zero torque. For steady state operation, the motor armature current is found to drop to zero at certain instances of time. At such instances, the voltage assumes a value that is

A. equal to the instantaneous value of the ac phase voltage

B. equal to the instantaneous value of the motor back emf

C. arbitrary

D. zero

Q2.00.2 MCQ 2000 1M

A thyristor, three phase, fully controlled converter feeds a dc load that draws a constant current. Then the input ac line current to the converter has

A. an rms value equal to the dc load current

B. an average value equal to the dc load current

C. a peak value equal to the dc load current

D. a fundamental frequency component, whose rms value is equal to the dc load current

Q2.00.3 MCQ 2000 5M

A single phase voltage source of magnitude V_s and frequency ω (rad/s) is connected to an inductance L through an antiparallel back-to-back pair of thyristors. The forward and reverse conducting thyristors are fired at an angle of $\alpha \geq \pi/2$ from the positive-going and negative-going zero crossings of the supply voltage respectively, in each cycle. Obtain an expression for the inductor current in each cycle for a given value of α. The voltage drop across the thyristors, when either of them is in conduction, may be assumed to be negligible.

Q2.99.1 MCQ 1999 1M

Resonant converters are basically used to

A. Generate large peak voltage

B. Reduce the switching losses

C. Eliminate harmonics

D. Convert a square wave into a sine wave

Q2.99.2 NAT 1999 5M

A three phase diode bridge is used to provide rectified output from a 400 V, 50 Hz, 3-phase supply to a R-L load with 10 Ω resistance and 300 mH inductance. Determine the (a) DC level of the output voltage (b) RMS value of the diode current (c) RMS value of the source current and (d) Apparent power drawn from the mains

Q2.98.1 MCQ 1998 2M

A 3-phase, fully controlled, converter is feeding power into a d.c. load at a constant current of 150 A. The rms current through each thyristor of the converter is:

A. 50 A **B.** 100 A **C.** $\frac{150\sqrt{2}}{\sqrt{3}}$ **D.** $\frac{150}{\sqrt{3}}$

Q2.98.2 MCQ 1998 2M

When the firing angle α of a single phase, fully controlled rectifier feeding constant d.c. current into a load is 30°, the displacement power factor of the rectifier is:

A. 1 **B.** 0.5 **C.** $\frac{1}{\sqrt{3}}$ **D.** $\frac{\sqrt{3}}{2}$

Q2.97.1 MCQ 1997 1M

In a dual converter, the circulating current

A. Allows smooth reversal of load current, but increases the response time

B. Does not allow smooth reversal of load current, but reduces the response time

C. Allows smooth reversal of load current with improved speed of response

D. Flows only if there is no interconnecting inductor

Q2.96.1 MCQ 1996 1M

In a 3-phase controlled bridge rectifier, as the overlap angle increases, the output DC voltage

A. Decreases **B.** Increases
C. Does not change **D.** Depends upon load inductance

Q2.95.1 NAT 1995 5M

The single phase half controlled ac to dc bridge converter of figure supplies a 10 Ω resistor in series with a 100 V back emf load. The firing angle of the thyristors is set to 60°.

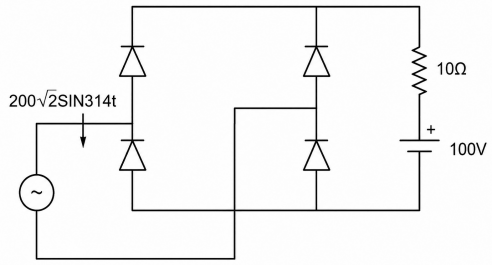


Fig. Q2.95.1

(i) Find the average current through the resistor.
(ii) What will be the new average current through the resistor, if a very large inductor is connected in series with the load?

Q2.95.2 NAT 1995 1M

A three phase ac-to-dc diode bridge rectifier is supplied from a three-phase, 440 V source. The rectifier supplies a purely resistive load. The average dc voltage across the load will be _____ V.

Q2.95.3 MCQ 1995 1M

A single phase diode bridge rectifier supplies a highly inductive load. The load current can be assumed to be ripple free. The ac supply side current waveform will be:

A. Sinusoidal
B. Constant dc
C. Square
D. Triangular

Q2.94.1 NAT 1994 1M

The output voltage of a six-pulse double star rectifier is the same as that of a three-phase half-wave rectifier. (True/False)

Q2.94.2 NAT 1994 1M

A line-commutated inverter changes dc voltage to ac voltage. (True/False)

Q2.94.3 MCQ 1994 1M

Referring to the figure, the type of load is

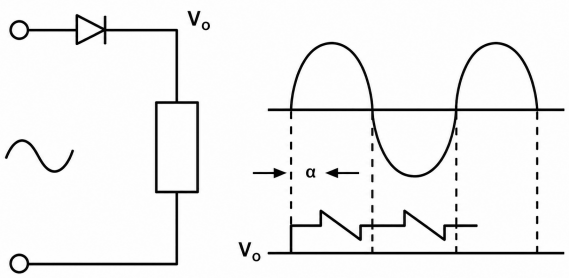


Fig. Q2.94.3

A. Inductive load
B. Resistive load
C. DC motor
D. Capacitive load

Q2.93.1 MCQ 1993 1M

When a line commutated converter operates in the inverter mode

A. It draws both real and reactive power from the A.C. supply

B. It delivers both real and reactive power to the A.C. supply

C. It delivers real power to the A.C. supply

D. It draws reactive power from the A.C. supply

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q2.26.1 32.3 to 32.5	Q2.26.2 1.04 to 1.14	Q2.25.1 A
Q2.25.2 A	Q2.25.3 A	Q2.24.1 0.940 to 0.970
Q2.24.2 113-116	Q2.23.1 39 to 40.5	Q2.22.1 800
Q2.22.2 0.43 to 0.47	Q2.22.3 17.65	Q2.21.1 389 to 391
Q2.20.1 C	Q2.20.2 4.51 to 5.10	Q2.19.1 C
Q2.19.2 81 to 82	Q2.19.3 0.75 to 0.8	Q2.18.1 C

Q2.18.2 16.90 to 17.70	Q2.18.3 90 to 90	Q2.17.1 220 to 230
Q2.17.2 0	Q2.17.3 C	Q2.17.4 0.71
Q2.16.1 C	Q2.16.2 9.21 to 9.22	Q2.16.3 6
Q2.16.4 57 to 58	Q2.15.1 B	Q2.15.2 C
Q2.15.3 61 to 62	Q2.14.1 23 to 23	Q2.14.2 69 to 70
Q2.14.3 0.74 to 0.82	Q2.14.4 16.6 to 17.4	Q2.14.5 B

Q2.13.1 C	Q2.12.1 D	Q2.11.1 B
Q2.11.2 C	Q2.11.3 B	Q2.11.4 B
Q2.10.1 A	Q2.09.1 C	Q2.08.1 A
Q2.08.2 C	Q2.08.3 B	Q2.08.4 B
Q2.07.1 C	Q2.07.2 C	Q2.07.3 C
Q2.07.4 A	Q2.07.5 D	Q2.07.6 D

Q2.06.1 D	Q2.06.2 C	Q2.06.3 A
Q2.06.4 C	Q2.05.1 B	Q2.05.2 B
Q2.04.1 C	Q2.03.1 A	Q2.03.2 A
Q2.02.1 C	Q2.02.2 B	Q2.02.3 C
Q2.01.1 A	Q2.01.2 A	Q2.00.1 B
Q2.00.2 C	Q2.00.3 *	Q2.99.1 B
Q2.99.2 *	Q2.98.1 D	Q2.98.2 D
Q2.97.1 C	Q2.96.1 A	Q2.95.1 7.406, 3.504
Q2.95.2 594.208	Q2.95.3 C	Q2.94.1 FALSE
Q2.94.2 TRUE	Q2.94.3 C	Q2.93.1 C

PrepFusion

UNIT

III

DC to DC Converters-Choppers

Topics

1. Buck Converter or Step Down Chopper with R/RL/RLE Load
2. Step Up Chopper or Boost Converter
3. Buck-Boost Converter
4. Fourier Series Analysis and Design of DC-DC Converters
5. Commutation Circuits

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 51	**MCQ:** 30	**MSQ:** 0	**NAT:** 21
1M: 14	**2M:** 33	**Descriptive:** 4	**Avg Weightage** ~3M	**Range:** 1M(2023)–6M(2013)

**See preface for how Avg Weightage and Range are calculated.*

Q3.26.1 MCQ 2026 2M

Consider the boost converter circuit shown. Assume that the semiconductor devices are ideal. In steady state, the inductor current rises linearly from 0 A to 6 A in the first 10 μs and then falls linearly from 6 A to 0 A in the next 10 μs of every switching cycle as shown. The load resistance R is 10 Ω and the capacitance C is 500 μF.

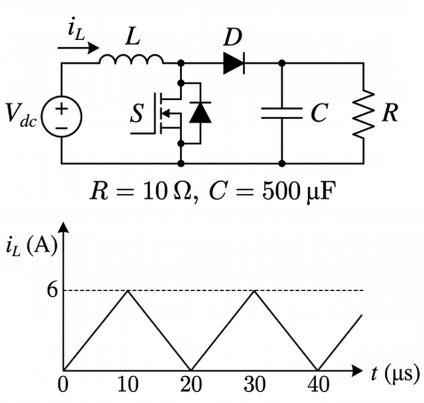


Fig. Q3.26.1

Neglect the ripple in the output voltage. What is the input voltage V_{dc}?

A. 10.0 V **B.** 15.0 V **C.** 7.5 V **D.** 12.5 V

Q3.25.1 NAT 2025 1M

In the circuit with ideal devices, the power MOSFET is operated with a duty cycle of 0.4 in a switching cycle with I = 10 A and V = 15 V. The power delivered by the current source, in W, is _____ (round off to the nearest integer).

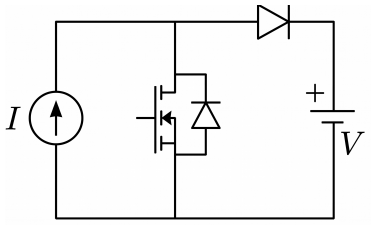


Fig. Q3.25.1

Q3.25.2 NAT 2025 2M

The steady state capacitor current of a conventional DC-DC buck converter, working in CCM, is shown in one switching cycle. If the input voltage is 30 V, the value of the inductor used, in mH, is _____ (round off to one decimal place).

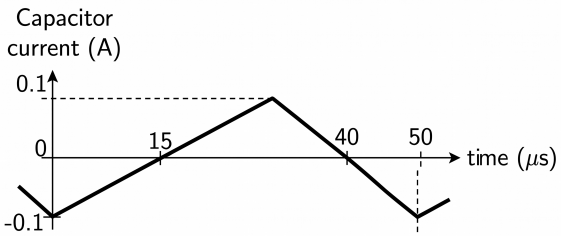


Fig. Q3.25.2

Q3.24.1 NAT 2024 1M

A forced commutated thyristorized step-down chopper is shown in the figure. Neglect the ON-state drop across the power devices. Assume that the capacitor is initially charged to 50 V with the polarity shown in the figure. The load current (I_L) can be assumed to be constant at 10 A. Initially, Th_M is ON and Th_A is OFF. The turn-off time available to Th_M in microseconds, when Th_A is triggered, is _____ (rounded off to the nearest integer).

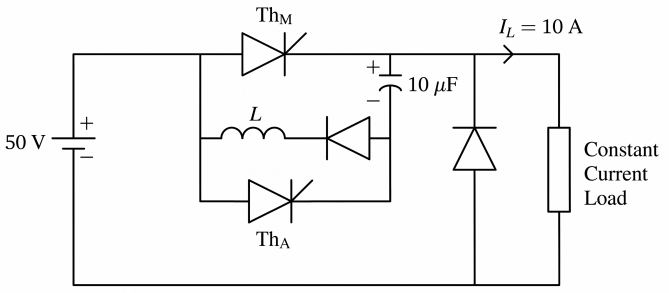


Fig. Q3.24.1

Q3.24.2 NAT 2024 2M

In the DC-DC converter shown in the figure, the current through the inductor is continuous. The switching frequency is 500 Hz. The voltage (V_o) across the load is assumed to be constant and ripple free. The peak inductor current in amperes is _____ (rounded off to the nearest integer).

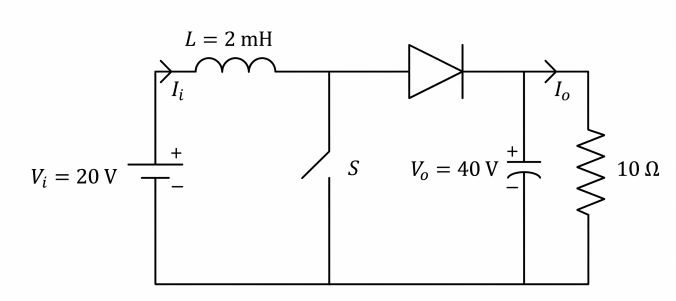


Fig. Q3.24.2

Q3.23.1 MCQ 2023 1M

The chopper circuit shown in figure (i) feeds power to a 5 A DC constant current source. The switching frequency of the chopper is 100 kHz. All the components can be assumed to be ideal. The gate signals of switches S_1 and S_2 are shown in figure (ii). Average voltage across the 5 A current source is

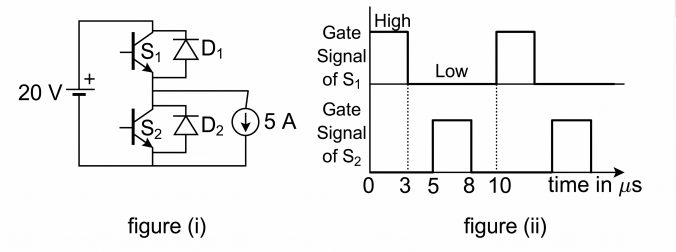


figure (i) figure (ii)

Fig. Q3.23.1

A. 10.0 *V* **B.** 6.0 *V* **C.** 12.0 *V* **D.** 20.0 *V*

Q3.22.1 NAT 2022 2M

The steady state current flowing through the inductor of a DC–DC buck–boost converter is given in the figure below. If the peak-to-peak ripple in the output voltage of the converter is 1 V, then the value of the output capacitor, in μF, is _____. *(round off to nearest integer)*

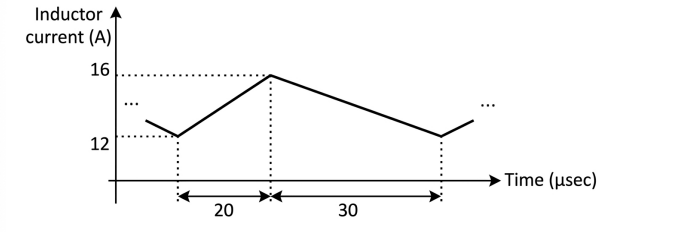


Fig. Q3.22.1

Q3.21.1 NAT 2021 2M

Consider the boost converter shown. Switch Q is operating at 25 kHz with a duty cycle of 0.6. Assume the diode and switch to be ideal. Under steady-state condition, the average resistance R_{in} as seen by the source is_____ Ω. (Round off to 2 decimal places.)

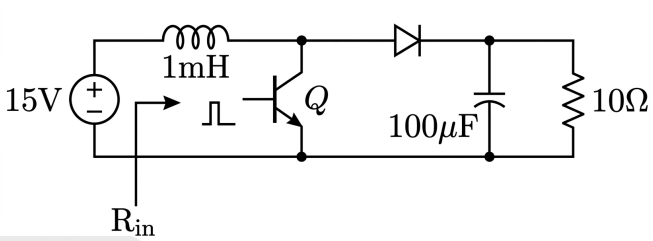


Fig. Q3.21.1

Q3.21.2 NAT 2021 2M

Consider the buck-boost converter shown. Switch Q is operating at 25 kHz and 0.75 duty-cycle. Assume diode and switch to be ideal. Under steady-state condition, the average current flowing through the inductor is _____ A.

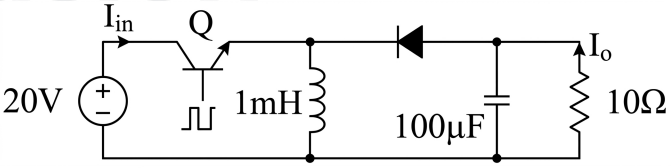


Fig. Q3.21.2

Q3.20.1 NAT 2020 2M

In the dc-dc converter circuit shown, switch Q is switched at a frequency of 10 kHz with a duty ratio of 0.6. All components of the circuit are ideal, and the initial current in the inductor is zero. Energy stored in the inductor in mJ (rounded off to 2 decimal places) at the end of 10 complete switching cycles is _____.

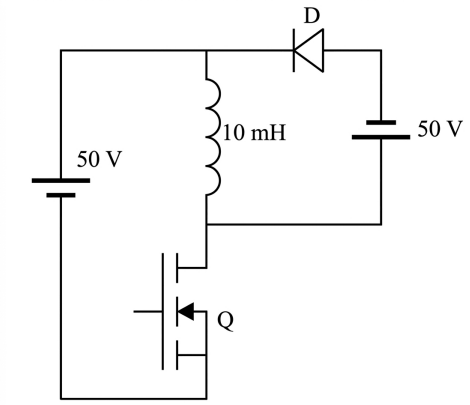


Fig. Q3.20.1

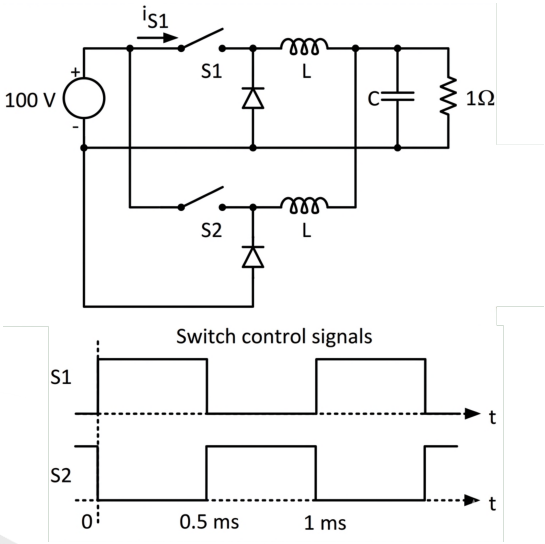


Fig. Q3.18.1

Q3.19.1 MCQ 2019 2M

A DC-DC buck converter operates in continuous conduction mode. It has 48 V input voltage, and it feeds a resistive load of 24 Ω. The switching frequency of the converter is 250 Hz. If switch-on duration is 1 ms, the load power is _____.

A. 6 W **B.** 12 W **C.** 24 W **D.** 48 W

Q3.19.2 NAT 2019 2M

In a DC-DC boost converter, the duty ratio is controlled to regulate the output voltage at 48 V. The input DC voltage is 24 V. The output power is 120 W. The switching frequency is 50 kHz. Assume ideal components and a very large output filter capacitor. The converter operates at the boundary between continuous and discontinuous conduction modes. The value of the boost inductor (in μH) is _____.

Q3.18.1 NAT 2018 2M

The figure shows two buck converters connected in parallel. The common input dc voltage for the converters has a value of 100 V. The converters have inductors of identical value. The load resistance is 1 Ω. The capacitor voltage has negligible ripple. Both converters operate in the continuous conduction mode. The switching frequency is 1 kHz, and the switch control signals are as shown. The circuit operates in the steady state. Assuming that the converters share the load equally, the average value of i_{S1}, the current of switch $S1$ (in Ampere), is _____ (up to 2 decimal places).

Q3.18.2 NAT 2018 2M

A dc to dc converter shown in the figure is charging a battery bank, $B2$ whose voltage is constant at 150 V. $B1$ is another battery bank whose voltage is constant at 50 V. The value of the inductor, L is 5 mH and the ideal switch, S is operated with a switching frequency of 5 kHz with a duty ratio of 0.4. Once the circuit has attained steady state and assuming the diode D to be ideal, the power transferred from $B1$ to $B2$ (in Watt) is _____ (up to 2 decimal places).

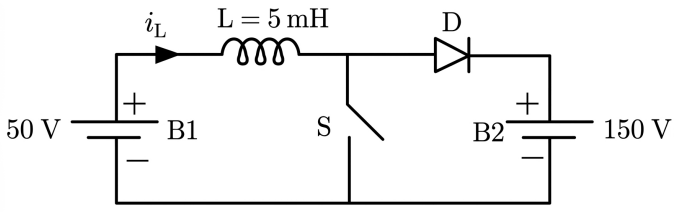


Fig. Q3.18.2

Q3.17.1 MCQ 2017 2M

The input voltage V_{DC} of the buck-boost converter shown below varies from 32 V to 72 V. Assume that all components are ideal, inductor current is continuous, and output voltage is ripple free. The range of duty ratio D of the converter for which the magnitude of the steady-state output voltage remains constant at 48 V is

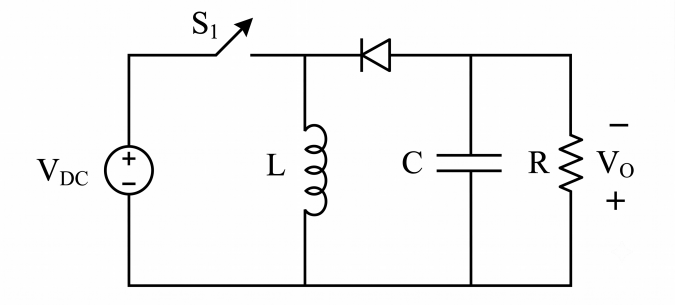


Fig. Q3.17.1

A. $\frac{2}{5} \le D \le \frac{3}{5}$ **B.** $\frac{2}{3} \le D \le \frac{3}{4}$

C. $0 \le D \le 1$ **D.** $\frac{1}{3} \le D \le \frac{2}{3}$

Q3.16.1 NAT 2016 1M

A buck converter, as shown in Figure (a) below, is working in steady state. The output voltage and the inductor current can be assumed to be ripple free. Figure (b) shows the inductor voltage v_L during a complete switching interval. Assuming all devices are ideal, the duty cycle of the buck converter is _____.

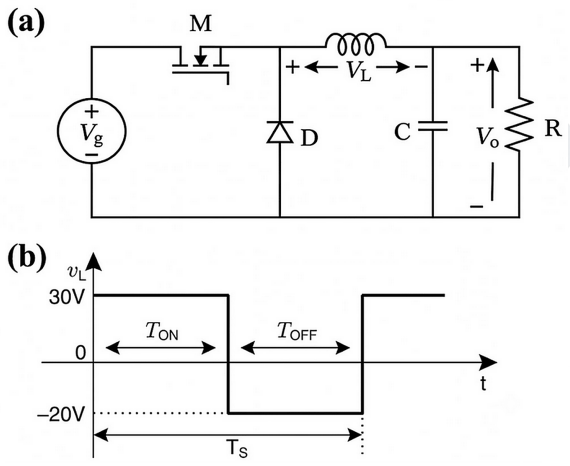


Fig. Q3.16.1

Q3.15.1 NAT 2015 1M

In the following chopper, the duty ratio of switch S is 0.4. If the inductor and capacitor are sufficiently large to ensure continuous inductor current and ripple-free capacitor voltage, the charging current (in Ampere) of the 5 V battery, under steady-state, is _____.

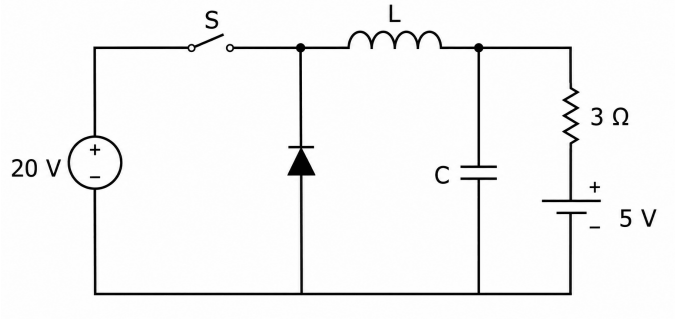


Fig. Q3.15.1

Q3.15.2 NAT 2015 2M

The circuit shown is meant to supply a resistive load R_L from two separate DC voltage sources. The switches S_1 and S_2 are controlled so that only one of them is ON at any instant. S_1 is turned on for 0.2 ms and S_2 is turned on for 0.3 ms in a 0.5 ms switching cycle time period. Assuming continuous conduction of the inductor current and negligible ripple on the capacitor voltage, the output voltage V_O (in Volt) across R_L is _____.

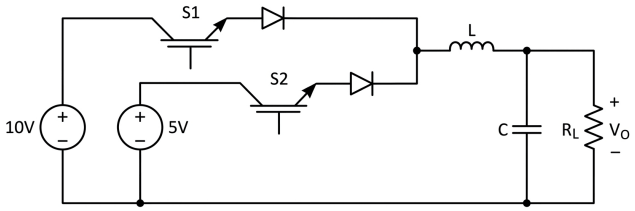


Fig. Q3.15.2

Q3.15.3 MCQ 2015 2M

A self commutating switch SW, operated at duty cycle δ is used to control the load voltage as shown in the figure

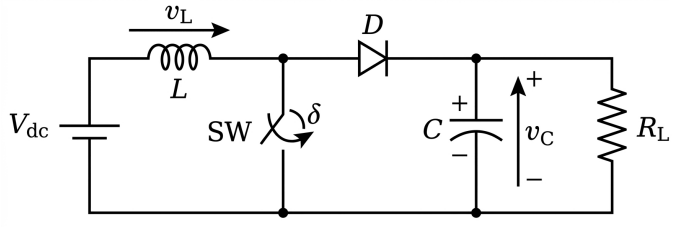


Fig. Q3.15.3

Under steady state operating conditions, the average voltage across the inductor and the capacitor respectively, are

A. $V_L = 0$ and $V_C = \frac{1}{1-\delta} V_{dc}$

B. $V_L = \frac{\delta}{2}V_{dc}$ and $V_C = \frac{1}{1-\delta}V_{dc}$

C. $V_L = 0$ and $V_C = \frac{\delta}{1-\delta}V_{dc}$

D. $V_L = \frac{\delta}{2}V_{dc}$ and $V_C = \frac{\delta}{1-\delta}V_{dc}$

Q3.15.4 NAT 2015 2M

A buck converter feeding a variable resistive load is shown in the figure. The switching frequency of the switch S is 100 kHz and the duty ratio is 0.6. The output voltage V_O is 36 V. Assume that all the components are ideal, and that the output voltage is ripple-free. The value of R (in Ohm) that will make the inductor current (i_L) just continuous is _____.

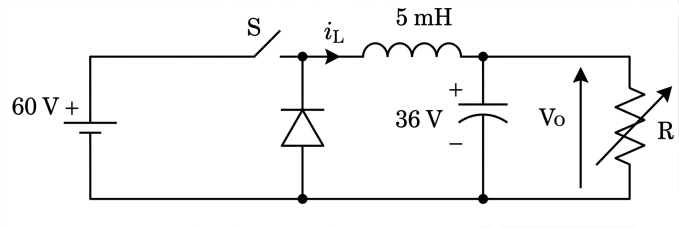


Fig. Q3.15.4

Q3.15.5 NAT 2015 2M

For the switching converter shown in the following figure, assume steady-state operation. Also assume that the components are ideal, the inductor current is always positive and continuous, and the switching period is T_s. If the voltage V_L is as shown, the duty cycle of the switch S is _____.

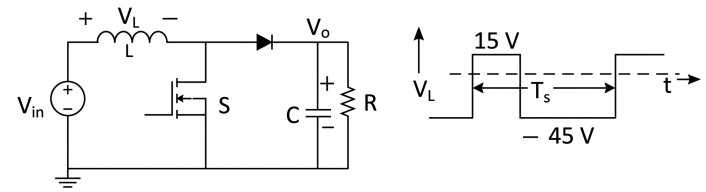


Fig. Q3.15.5

Q3.14.1 NAT 2014 2M

Figure (i) shows the circuit diagram of a chopper. The switch S in the circuit in figure (i) is switched such that the voltage v_D across the diode has the wave shape as shown in figure (ii). The capacitance C is large so that the voltage across it is constant. If switch S and the diode are ideal, the peak to peak ripple (in A) in the inductor current is _____

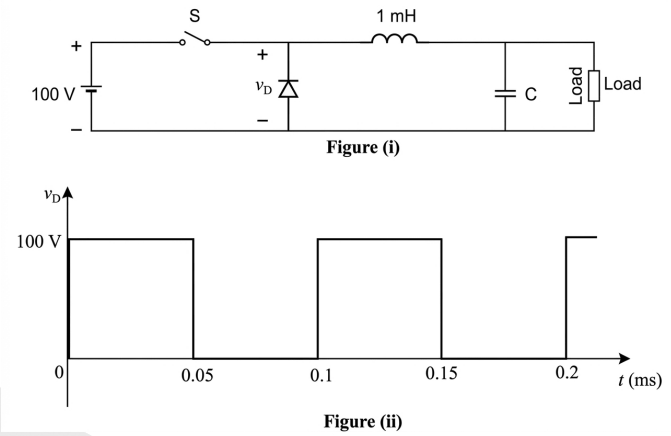


Fig. Q3.14.1

Q3.14.2 NAT 2014 1M

A step-up chopper is used to feed a load at 400 V dc from a 250 V dc source. The inductor current is continuous. If the 'off' time of the switch is 20 μs, the switching frequency of the chopper in kHz is _____.

Q3.13.1 MCQ 2013 2M

The separately excited dc motor in the figure below has a rated armature current of 20 A and a rated armature voltage of 150 V. An ideal chopper switching at 5 kHz is used to control the armature voltage. If $L_a = 0.1$ mH, $R_a = 1\ \Omega$, neglecting armature reaction, the duty ratio of the chopper to obtain 50% of the rated torque at the rated speed and the rated field current is _____.

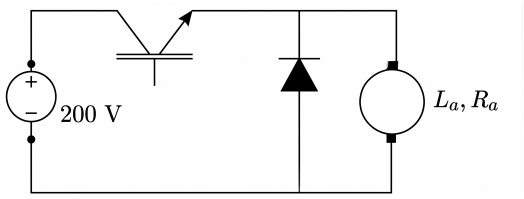


Fig. Q3.13.1

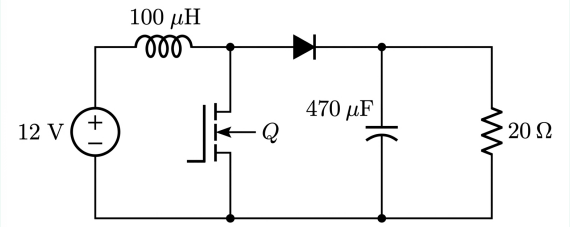
A. 0.4 **B.** 0.5 **C.** 0.6 **D.** 0.7

Common Data Questions

Common Data for Questions Q3.13.2 and Q3.13.3

In the figure shown below, the chopper feeds a resistive load from a battery source. MOSFET Q

is switched at 250 kHz, with a duty ratio of 0.4. All elements of the circuit are assumed to be ideal.



Q3.13.2 MCQ 2013 2M

The average source current in Amps in steady-state is

A. 3/2 **B.** 5/3 **C.** 5/2 **D.** 15/4

Q3.13.3 MCQ 2013 2M

The PEAK-TO-PEAK source current ripple in Amps is

A. 0.96 **B.** 0.144 **C.** 0.192 **D.** 0.288

Q3.12.1 MCQ 2012 2M

In the circuit shown, an ideal switch S is operated at 100 kHz with a duty ratio of 50 %. Given that Δi_c is 1.6 A peak-to-peak and I_0 is 5 A dc, the peak current in S is

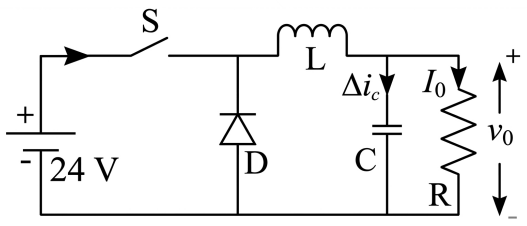


Fig. Q3.12.1

A. 6.6A **B.** 5.0A **C.** 5.8A **D.** 4.2A

Q3.11.1 MCQ 2011 2M

A voltage commutated chopper circuit, operated at 500 Hz, is shown below.

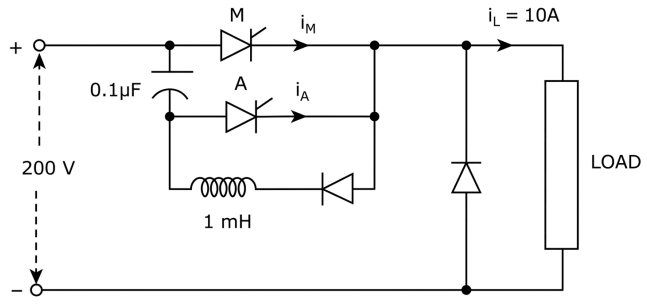


Fig. Q3.11.1

If the maximum value of load current is 10 A, then the maximum current through the main (M) and auxiliary (A) thyristors will be

A. $i_{M_{max}} = 12$ A and $i_{A_{max}} = 10$ A

B. $i_{M_{max}} = 12$ A and $i_{A_{max}} = 2$ A

C. $i_{M_{max}} = 10$ A and $i_{A_{max}} = 12$ A

D. $i_{M_{max}} = 10$ A and $i_{A_{max}} = 8$ A

Q3.10.1 MCQ 2010 1M

The power electronic converter shown in the figure has a single-pole double-throw switch. The pole P of the switch is connected alternately to throws A and B. The converter shown is a

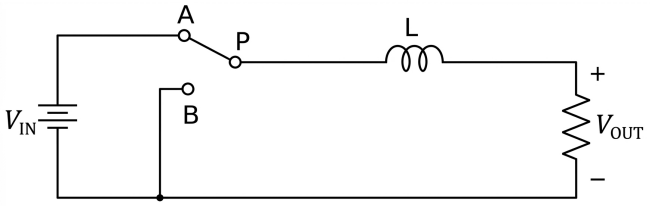


Fig. Q3.10.1

A. step-down chopper (buck converter)

B. half-wave rectifier

C. step-up chopper (boost converter)

D. full-wave rectifier

Q3.09.1 MCQ 2009 2M

In the chopper circuit shown, the main thyristor (T_M) is operated at a duty ratio of 0.8 which is much larger than the commutation interval. If the maximum allowable reapplied $\frac{dv}{dt}$ on T_M is 50 V/μs, what should be the theoretical minimum value of C_1? Assume current ripple through L_0 to be negligible.

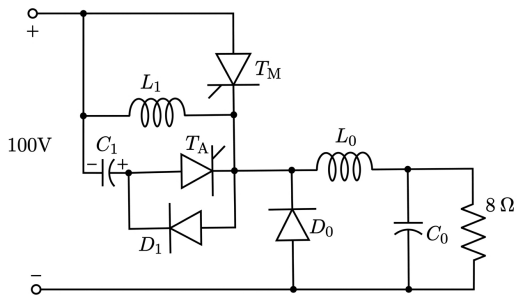


Fig. Q3.09.1

A. 0.2 μF **B.** 0.02 μF **C.** 2 μF **D.** 20 μF

Q3.08.1 MCQ 2008 2M

In the circuit shown in the figure, the switch is operated at a duty cycle of 0.5. A large capacitor is connected across the load. The inductor current is assumed to be continuous.

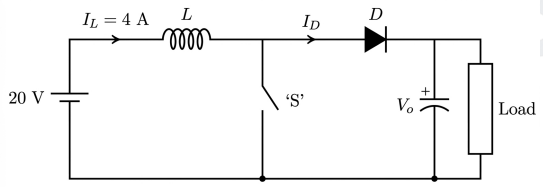


Fig. Q3.08.1

The average voltage across the load and the average current through the diode will respectively be

A. 10 V, 2 A **B.** 10 V, 8 A
C. 40 V, 2 A **D.** 40 V, 8 A

Q3.07.1 MCQ 2007 2M

The circuit in the figure is a current commutated dc – dc chopper where, Th_M is the main SCR and Th_{AUX} is the auxiliary SCR. The load current is constant at 10 A. Th_M is ON. Th_{AUX} is triggered at $t = 0$. Th_M is turned OFF between

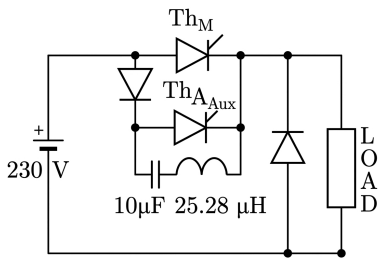


Fig. Q3.07.1

A. $0\,\mu s < t \le 25\,\mu s$ **B.** $25\,\mu s < t \le 50\,\mu s$
C. $50\,\mu s < t \le 75\,\mu s$ **D.** $75\,\mu s < t \le 100\,\mu s$

Common Data Questions

Common Data for Questions Q3.06.1 and Q3.06.2
A voltage commutated chopper operating at 1 kHz is used to control the speed of dc motor as shown in figure. The load current is assumed to be constant at 10 A

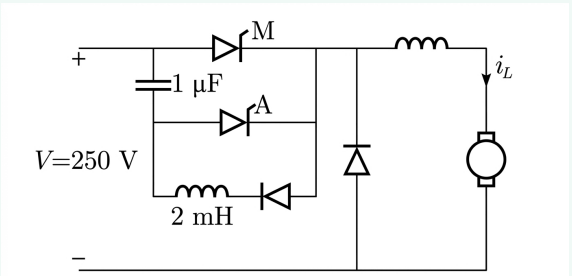


Fig. Q3.07.1

Q3.06.1 MCQ 2006 2M

The minimum time in μsec for which the SCR M should be ON is

A. 280 **B.** 140 **C.** 70 **D.** 0

Q3.06.2 MCQ 2006 2M

The average output voltage of the chopper will be

A. 70 V **B.** 47.5 V **C.** 35 V **D.** 0 V

Q3.05.1 MCQ 2005 2M

The given figure shows a step-down chopper switched at 1 kHz with a duty ratio $D = 0.5$. The peak-peak ripple in the load current is close to

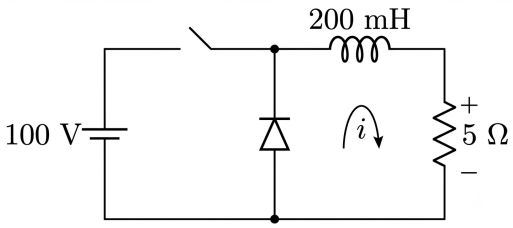


Fig. Q3.05.1

A. 10 A **B.** 0.5 A **C.** 0.125 A **D.** 0.25 A

Q3.04.1 MCQ 2004 2M

Figure shows a chopper operating from a 100 V dc input. The duty ratio of the main switch S is 0.8. The load is sufficiently inductive so that the load current is ripple free. The average current through the diode D under steady state is

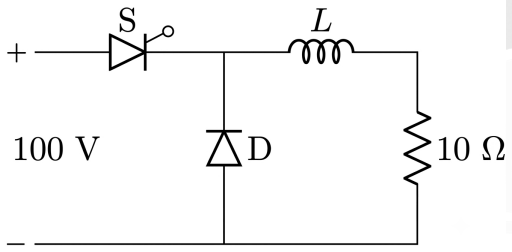


Fig. Q3.04.1

A. 1.6 A **B.** 6.4 A **C.** 8.0 A **D.** 10.0 A

Q3.04.2 MCQ 2004 2M

Figure shows a chopper. The device S_1 is the main switching device. S_2 is the auxiliary commutation device. S_1 is rated for 400 V, 60 A. S_2 is rated for 400 V, 30 A. The load current is 20 A. The main device operates with a duty ratio of 0.5. The peak current through S_1 is

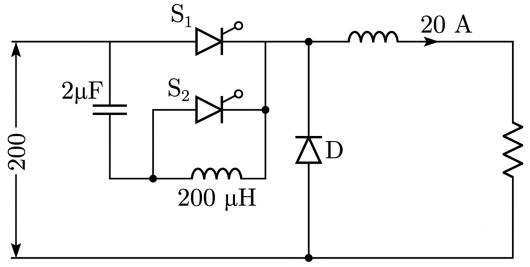


Fig. Q3.04.2

A. 10 A **B.** 20 A **C.** 30 A **D.** 40 A

Q3.03.1 MCQ 2003 2M

A chopper is employed to charge a battery as shown in figure. The charging current is 5 A. The duty ratio is 0.2. The chopper output voltage is also shown in the figure. The peak to peak ripple current in the charging current is

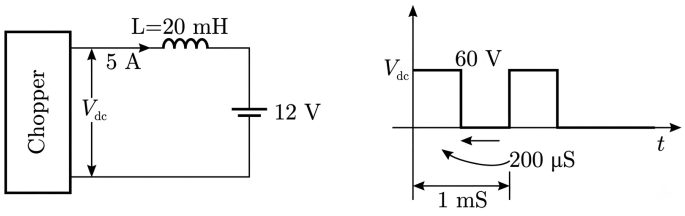


Fig. Q3.03.1

A. 0.48 A **B.** 1.2 A **C.** 2.4 A **D.** 1 A

Q3.02.1 MCQ 2002 1M

A step down chopper is operated in the continuous conduction mode in steady state with a constant duty ratio D. If V_0 is the magnitude of the dc output voltage and if V_s is the magnitude of the dc input voltage, the ratio $\frac{V_0}{V_s}$ is given by

A. D **B.** $1 - D$ **C.** $\frac{1}{1-D}$ **D.** $\frac{D}{1-D}$

Q3.02.2 MCQ 2002 2M

In the chopper circuit shown in Fig. P2.22, the input DC voltage has a constant value V_s. The output voltage V_o is assumed ripple-free. The switch S is operated with a switching period T and duty ratio D. What is the value of D at the boundary between continuous and discontinuous conduction of the inductor current i_L?

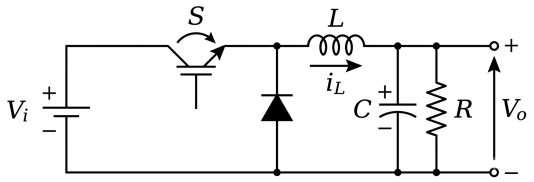


Fig. Q3.02.2

A. $D = 1 - \frac{V_s}{V_o}$ **B.** $D = \frac{2L}{RT}$

C. $D = 1 - \frac{2L}{RT}$ **D.** $D = \frac{RT}{L}$

Q3.02.3 MCQ 2002 5M

In the figure, the ideal switch S is switched on and off with a switching frequency $f = 10$ kHz. The switching time period is $T = t_{ON} + t_{OFF} = 100\ \mu s$. The circuit is operated in steady state at the boundary of continuous and discontinuous conduction, so that the inductor current i is as shown in Figure. Find

(a) The on-time t_{ON} of the switch.

(b) The value of the peak current I_p.

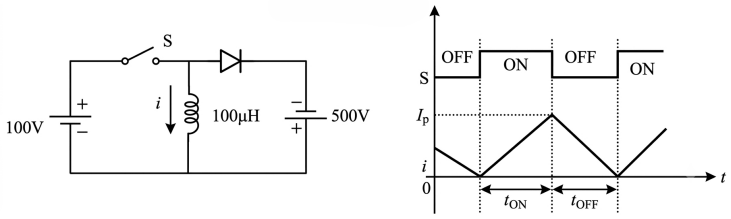


Fig. Q3.02.3

Q3.02.4 MCQ 2002 5M

In the circuit shown in the figure, the source I is a dc current source. The switch S is operated with a time period T and a duty ratio D. You may assume that the capacitance C has a finite value which is large enough so that the voltage. V_c has negligible ripple. Calculate the following under steady state conditions, in terms of D, I and R.

(a) The voltage V_c, with the polarity shown in Figure.

(b) The average output voltage V_o, with the polarity shown.

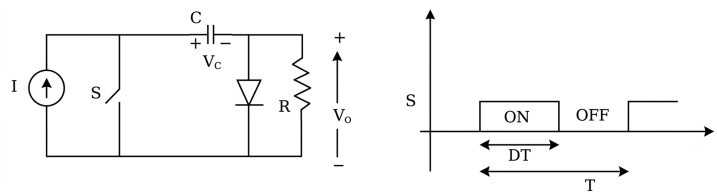


Fig. Q3.02.4

Q3.01.1 MCQ 2001 5M

A voltage commutated thyristor chopper circuit is shown in the figure. The chopper is operated at 500 Hz with 50% duty ratio. The load takes a constant current of 20 A.

(a) Evaluate the circuit turn off time for the main thyristor Th_1.

(b) Calculate the value of inductor L, if the peak current through the main thyristor Th_1 is limited to 180% of the load current.

(c) Calculate the maximum instantaneous output voltage of the chopper.

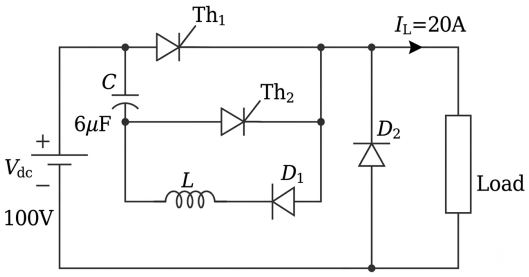


Fig. Q3.01.1

Q3.00.1 MCQ 2000 2M

A step down chopper operates from a dc voltage source V_s, and feeds a dc motor armature with a back emf E_b. From oscilloscope traces, it is found that the current increases for time t_r, falls to zero over time t_f, and remains zero for time t_0 in every chopping cycle.
The average dc voltage across the freewheeling diode is:

A. $\dfrac{V_s t_r}{t_r + t_f + t_0}$

B. $\dfrac{V_s t_r + E_b t_f}{t_r + t_f + t_0}$

C. $\dfrac{V_s t_r + E_b t_0}{t_r + t_f + t_0}$

D. $\dfrac{V_s t_r + E_b (t_f + t_0)}{t_r + t_f + t_0}$

Q3.99.1 MCQ 1999 1M

Resonant converters are primarily used to

A. Generate large peaky voltage

B. Reduce the switching losses

C. Eliminate harmonics

D. Convert a square wave into a sine wave

Q3.96.1 MCQ 1996 1M

In a thyristor dc chopper, which type of commutation results in best performance?

A. Voltage commutation

B. Current commutation

C. Load commutation

D. Supply commutation

Q3.95.1 NAT 1995 1M

Consider the chopper circuit of figure. The chopper operates at 400Hz and 50% duty cycle. The load current remains almost ripple free at 10A. Assuming the input voltage to be 200V and the devices to be ideal, the turn-off time available to the thyristor ThM is _____μs.

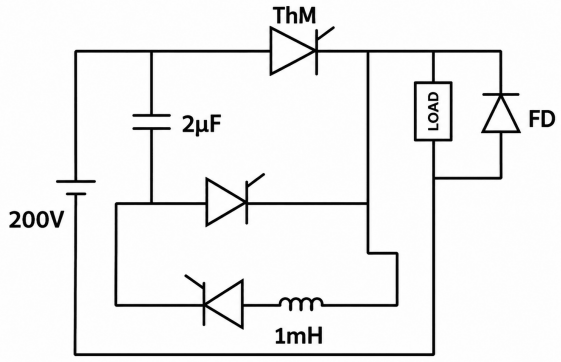


Fig. Q3.95.1

Q3.95.2 MCQ 1995 1M

A dc to dc transistor chopper supplied from a fixed voltage dc source feeds a fixed resistive-inductive load and a free-wheeling diode. The chopper operates at 1 kHz and 50% duty cycle. Without changing the value of the average dc current through the load, if it is desired to reduce the ripple content of load current, the control action needed will be:

A. Increase the chopper frequency keeping its duty cycle constant

B. Increase the chopper frequency and duty cycle in equal ratio

C. Decrease only the chopper frequency

D. Decrease only the duty cycle.

Q3.93.1 MCQ 1993 1M

A chopper operating at a fixed frequency is feeding an R-L load. As the duty ratio of the chopper is increased from 25% to 75%, the ripple in the load current

A. Remains constant

B. Decreases, reaches a minimum at 50% duty ratio and then increases

C. Increases, reaches a maximum at 50% duty ratio and then decreases

D. Keeps on increasing as the duty ratio is increased

Q3.92.1 NAT 1992 1M

In the circuit shown in figure, L is large and the average value of i is 100 A. The supply voltage is

$$e(t) = \sqrt{2} \times 200 \sin(314t) \text{ V}.$$

The thyristor is gated in the _____ half cycle of e at a delay angle α equal to _____.

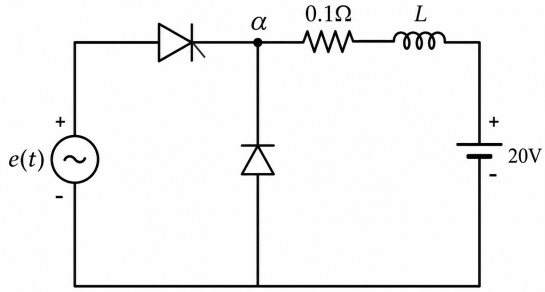


Fig. Q3.92.1

Q3.91.1 NAT 1991 5M

Figure shows the circuit schematic of a chopper driven, separately excited d.c. motor. The single-pole double-throw switch operates with a switching period (T_{ON}/T_s) is 0.2. The motor may be assumed lossless, with an armature inductance of 10 mH. The motor draws an average current of 20A at a constant back emf of 80V, under steady state.

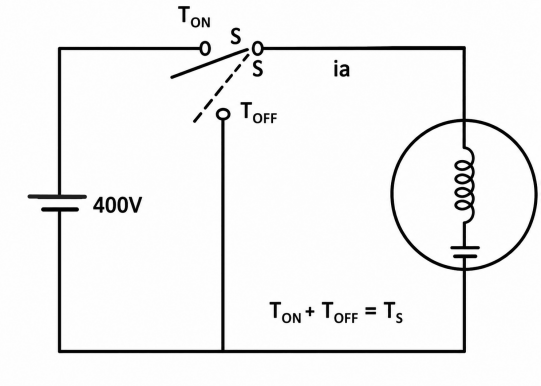


Fig. Q3.91.1

1. Sketch and label the voltage waveform, $v_0(t)$, of the chopper for one switching period.

2. Sketch and label the motor current, $i_a(t)$, for one switching period.

3. Evaluate the peak-to-peak current ripple of the motor.

Q3.15.3 A	Q3.15.4 2480 to 2520	Q3.15.5 0.75
Q3.14.1 2.49 to 2.51	Q3.14.2 31.0 to 31.5	Q3.13.1 D
Q3.13.2 B	Q3.13.3 C	Q3.12.1 C
Q3.11.1 A	Q3.10.1 A	Q3.09.1 A
Q3.08.1 C	Q3.07.1 C	Q3.06.1 B
Q3.06.2 B	Q3.05.1 C	Q3.04.1 A
Q3.04.2 D	Q3.03.1 A	Q3.02.1 A
Q3.02.2 C	Q3.02.3 *	Q3.02.4 *
Q3.01.1 *	Q3.00.1 C	Q3.99.1 B
Q3.96.1 A	Q3.95.1 40	Q3.95.2 A
Q3.93.1 C	Q3.92.1 POSITIVE, 109.485	Q3.91.1 *

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q3.26.1 C	Q3.25.1 90 to 90	Q3.25.2 1.7 to 1.9
Q3.24.1 50 to 50	Q3.24.2 13	Q3.23.1 B
Q3.22.1 168	Q3.21.1 1.55 to 1.65	Q3.21.2 24 to 24
Q3.20.1 4.95 to 5.05	Q3.19.1 A|C	Q3.19.2 24 to 24
Q3.18.1 11.5 to 13.5	Q3.18.2 12.00 to 12.00	Q3.17.1 A
Q3.16.1 0.39 to 0.41	Q3.15.1 1	Q3.15.2 7

UNIT

IV

DC to AC Converter- Inverter

Topics

1. Single-Phase Half Bridge Voltage Source Inverter
2. Single-Phase Full Bridge Voltage Source Inverter
3. Three-Phase Voltage Source Inverter in $180°$ Mode
4. Three-Phase Voltage Source Inverter in $120°$ Mode
5. Voltage Source Inverter with single and multiple PWM
6. Single-Phase Current Source Inverter
7. Three-Phase Current Source Inverter
8. Sinusoidal Pulse Width Modulation (SPWM)

PrepFusion

Unit Snapshot*

Stream: EE	**Questions:** 37	**MCQ:** 24	**MSQ:** 0	**NAT:** 13
1M: 14	**2M:** 22	**Descriptive:** 1	**Avg Weightage** ~2M	**Range:** 1M(2020)–4M(2022)

**See preface for how Avg Weightage and Range are calculated.*

Q4.26.1 MCQ 2026 2M

Consider the single-phase voltage source inverter circuit feeding an inductive load (L). Assume that the power MOSFET switches are ideal. S_1 and S_2 are switched on during the first 10 μs, and S_3 and S_4 are switched on during the next 10 μs in a switching cycle. The switches in the same leg are thus switched in a complementary fashion. Neglect the dead time. The waveform of the inductor current (i_L) in the steady state is triangular with a peak value of 5 A as shown.

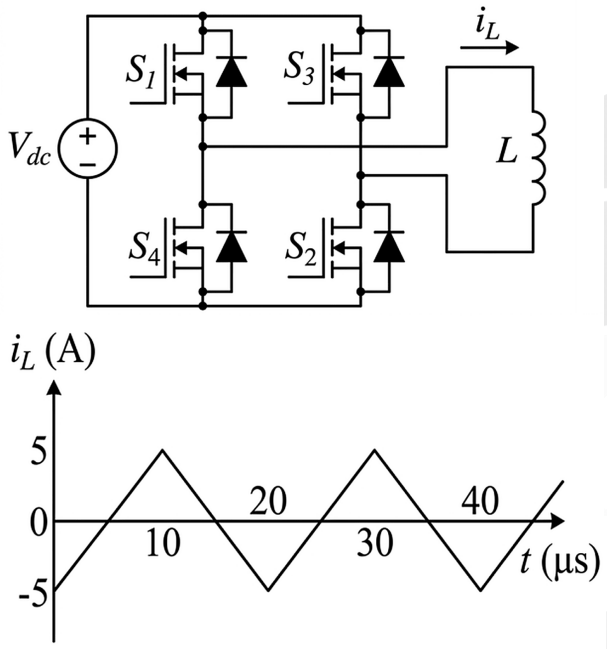


Fig. Q4.26.1

The rms value of the current through the switch S_1 is:

A. 2.88 A **B.** 2.04 A **C.** 3.54 A **D.** 2.50 A

Q4.25.1 MCQ 2025 2M

The 3-phase modulating waveforms ($v_a(t)$, $v_b(t)$ and $v_c(t)$), used in sinusoidal PWM in a Voltage Source Inverter (VSI) are

$$v_a(t) = 0.8\sin(\omega t)\ \text{V}$$

$$v_b(t) = 0.8\sin\left(\omega t - \frac{2\pi}{3}\right)\ \text{V}$$

$$v_c(t) = 0.8\sin\left(\omega t + \frac{2\pi}{3}\right)\ \text{V}$$

where $\omega = 2\pi \times 40$ rad/s is the fundamental frequency. The modulating waveforms are compared with a 10 kHz triangular carrier whose magnitude varies between +1 and −1. The VSI has a DC link voltage of 600 V and feeds a star connected motor. The per phase fundamental RMS motor voltage, in volts, is closest to

A. 169.71 **B.** 300.00 **C.** 424.26 **D.** 212.13

Q4.24.1 NAT 2024 2M

A single-phase full bridge voltage source inverter (VSI) feeds a purely inductive load. The inverter output voltage is a square wave in 180° conduction mode. The fundamental frequency of the output voltage is 50 Hz. If the DC input voltage of the inverter is 100 V and the value of the load inductance is 20 mH, the peak-to-peak load current in amperes is _____ (rounded off to the nearest integer).

Q4.22.1 NAT 2022 2M

Consider an ideal full-bridge single-phase DC-AC inverter with a DC bus voltage magnitude of 1000 V. The inverter output voltage $v(t)$ shown below, is obtained when diagonal switches of the inverter are switched with 50% duty cycle. The inverter feeds a load with a sinusoidal current given by,

$$i(t) = 10\sin\left(\omega t - \frac{\pi}{3}\right)\ \text{A, where } \omega = \frac{2\pi}{T}.$$

The active power, in watts, delivered to the load is _____ *(round off to nearest integer)*

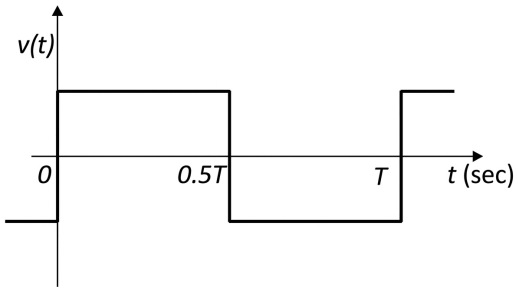


Fig. Q4.22.1

Q4.22.2 NAT 2022 2M

A 3-phase grid-connected voltage source converter with DC link voltage of 1000 V is switched using sinusoidal Pulse Width Modulation (PWM) technique. If the grid phase current is 10 A and the 3-phase complex power supplied by the converter is given by $(-4000 - j3000)$ VA, then the modulation index used in sinusoidal PWM is _____. *(round off to two decimal places)*

Q4.21.1 NAT 2021 2M

A single-phase full-bridge inverter fed by a 325 V DC produces a symmetric quasi-square waveform across 'ab' as shown. To achieve a modulation index of 0.8, the angle θ expressed in degrees should be _____. (Round off to 2 decimal places.)

(Modulation index is defined as the ratio of the peak of the fundamental component of V_{ab} to the applied DC value.)

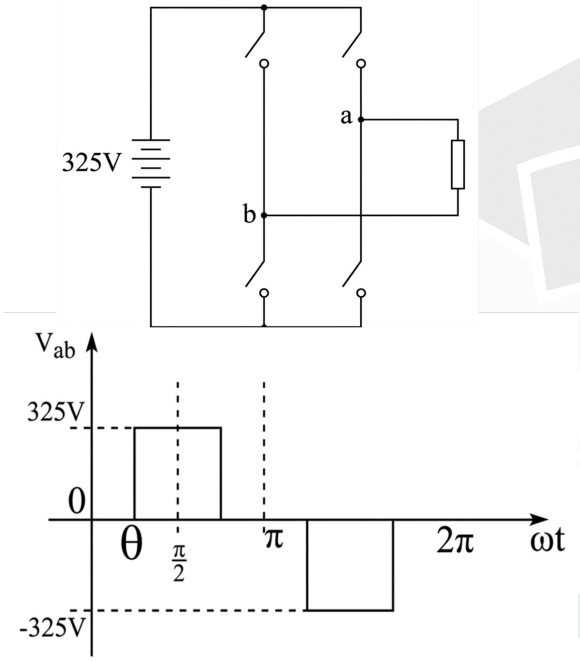


Fig. Q4.21.1

Q4.20.1 NAT 2020 1M

A single-phase inverter is fed from a 100 V dc source and is controlled using a quasi-square wave modulation scheme to produce an output waveform, $v(t)$, as shown. The angle σ is adjusted to entirely eliminate the 3rd harmonic component from the output voltage. Under this condition, for $v(t)$, the magnitude of the 5th harmonic component as a percentage of the magnitude of the fundamental component is _____ *(rounded off to 2 decimal places).*

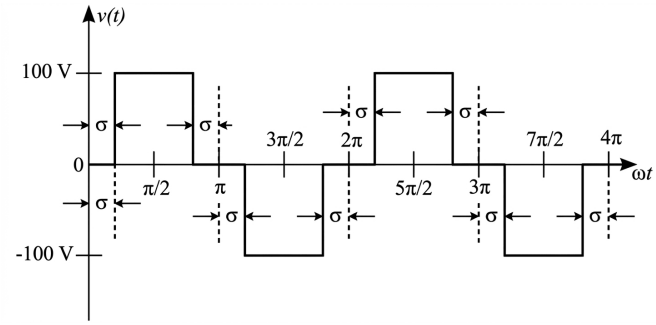


Fig. Q4.20.1

Q4.19.1 NAT 2019 1M

The output voltage of a single-phase full bridge voltage source inverter is controlled by unipolar PWM with one pulse per half cycle. For the fundamental rms component of output voltage to be 75% of DC voltage, the required pulse width in degrees (round off up to one decimal place) is _____.

Q4.17.1 NAT 2017 1M

In the converter circuit shown below, the switches are controlled such that the load voltage $v_o(t)$ is a 400 Hz square wave

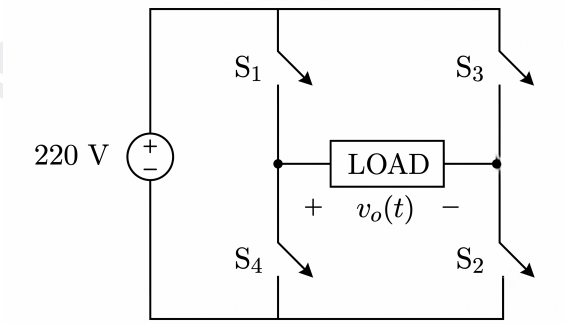


Fig. Q4.17.1

The RMS value of the fundamental component of $v_o(t)$ in volts is _____.

Q4.17.2 NAT 2017 1M

A 3-phase voltage source inverter is supplied from a 600 V DC source as shown in the figure below. For a star connected resistive load of 20 Ω per phase, the load power for 120° device conduction, in kW, is _____.

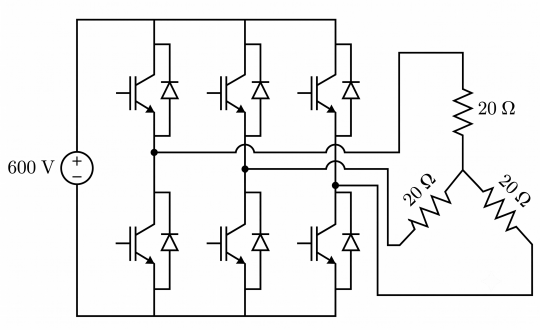


Fig. Q4.17.2

Q4.16.1 NAT 2016 2M

The switches T1 and T2 in Figure (a) are switched in a complementary fashion with sinusoidal pulse width modulation technique. The modulating voltage $v_m(t) = 0.8\sin(200\pi t)$ V and the triangular carrier voltage (v_c) are shown in Figure (b). The carrier frequency is 5 kHz. The peak value of the 100 Hz component of the load current (i_L), in ampere, is _____.

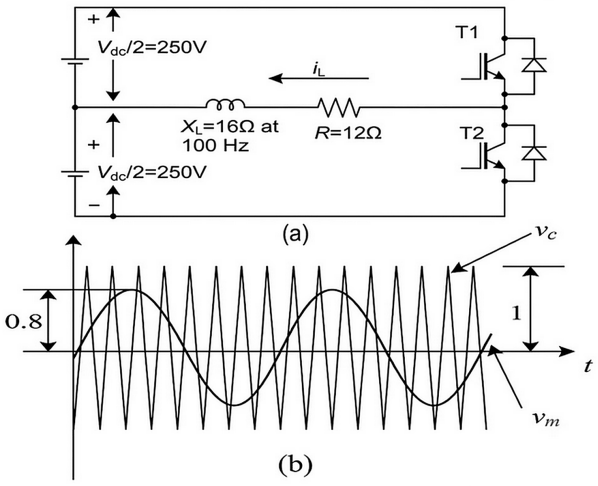


Fig. Q4.16.1

Q4.16.2 MCQ 2016 2M

A single-phase full-bridge voltage source inverter (VSI) is fed from a 300 V battery. A pulse of 120° duration is used to trigger the appropriate devices in each half-cycle. The rms value of the fundamental component of the output voltage, in volts, is

A. 234 **B.** 245 **C.** 300 **D.** 331

Q4.15.1 NAT 2015 2M

The single-phase full-bridge voltage source inverter (VSI), shown in figure, has an output frequency of 50 Hz. It uses unipolar pulse width modulation with switching frequency of 50 kHz and modulation index of 0.7. For V_{in} = 100 V DC, L = 9.55 mH, C = 63.66 μF, and R = 5 Ω, the amplitude of the fundamental component in the output voltage V_o (in Volt) under steady-state is _____.

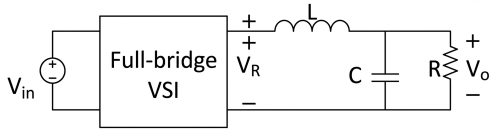


Fig. Q4.15.1

Q4.14.1 NAT 2014 2M

The figure shows one period of the output voltage of an inverter. α should be chosen such that $60° < \alpha < 90°$. If rms value of the fundamental component is 50 V, then α in degree is _____

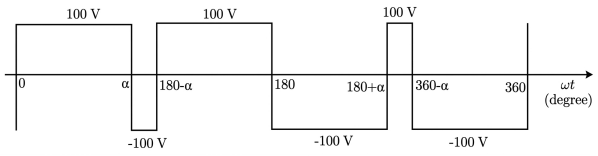


Fig. Q4.14.1

Q4.14.2 MCQ 2014 2M

A single-phase voltage source inverter shown in figure is feeding power to a load. The triggering pulses of the devices are also shown in the figure.

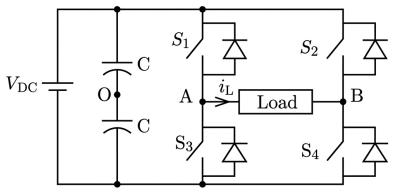


Fig. Q4.14.2

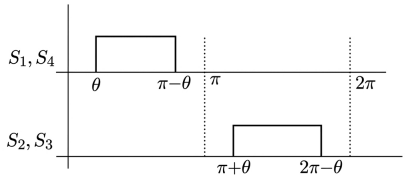
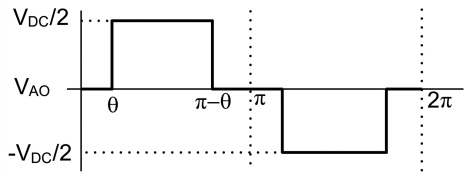
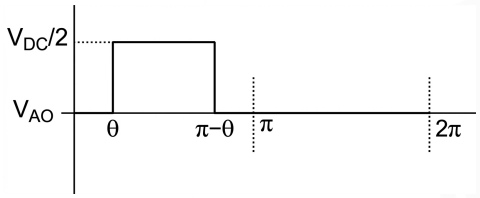
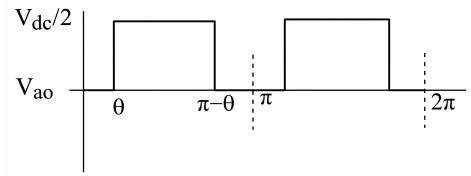


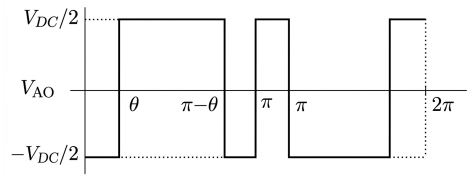
Fig. Q4.14.2

If the load current is sinusoidal and is zero at $0, \pi, 2\pi, \ldots,$ the node voltage V_{AO} has the waveform

A.


B.


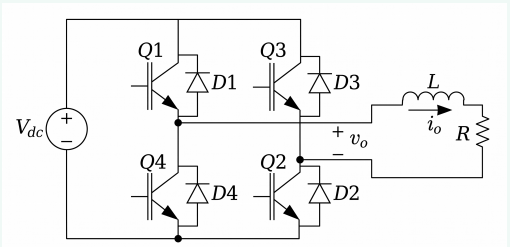
C.


D.


Common Data Questions

Common Data for Questions Q4.13.1 and Q4.13.2

The Voltage Source Inverter (VSI) shown in the figure below is switched to provide a 50 Hz, square-wave ac output voltage (v_o) across an R-L load. Reference polarity of v_o and reference direction of the output current i_o are indicated in the figure. It is given that $R = 3\,\Omega$, $L = 9.55$ mH.



Q4.13.1 MCQ 2013 1M

In the interval when $v_o < 0$ and $i_o > 0$, the pair of devices which conducts the load current is

A. Q_1, Q_2 **B.** Q_3, Q_4 **C.** D_1, D_2 **D.** D_3, D_4

Q4.13.2 MCQ 2013 1M

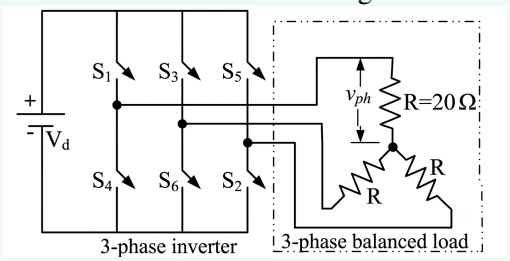
Appropriate transition i.e., Zero Voltage Switching (ZVS)/ Zero Current Switching (ZCS) of the IGBTs during turn-on/turn-off is

A. ZVS during turn-OFF **B.** ZVS during turn-ON
C. ZCS during turn-OFF **D.** ZCS during turn-ON

Common Data Questions

Common Data for Questions Q4.12.1 and Q4.12.2

In the 3-phase inverter circuit shown, the load is balanced and the gating scheme is 180°-conduction mode. All the switching devices are ideal.



Q4.12.1 MCQ 2012 2M

The rms value of load phase voltage is

A. 106.1 V **B.** 141.4 V **C.** 212.2 V **D.** 282.8 V

Q4.12.2 MCQ 2012 2M

If the dc bus voltage $V_d = 300$ V, the power consumed by the 3-phase load is

A. 1.5 kW **B.** 2.0 kW **C.** 2.5 kW **D.** 3.0 kW

Q4.11.1 MCQ 2011 1M

A three-phase current source inverter used for the speed control of an induction motor is to be realized using MOSFET switches as shown below. Switches S_1 to S_6 are identical switches.

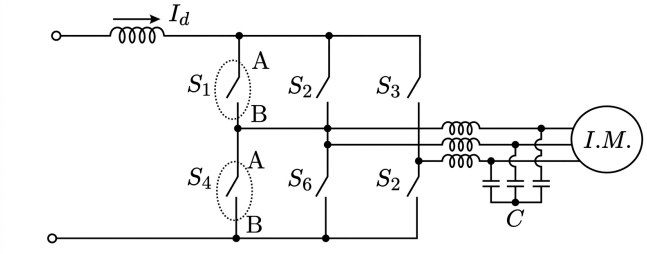
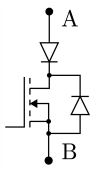
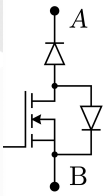
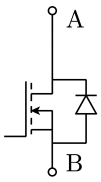


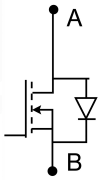
Fig. Q4.11.1

The proper configuration for realizing switches S_1 to S_6 is

A.


B.


C.


D.


Q4.09.1 MCQ 2009 2M

The Current Source Inverter shown in figure is operated by alternately turning on thyristor pairs (T_1, T_2) and (T_3, T_4). If the load is purely resistive, the theoretical maximum output frequency obtainable will be

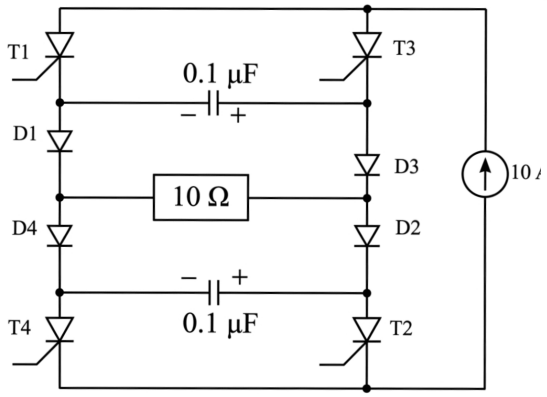


Fig. Q4.09.1

A. 125 kHz **B.** 250 kHz **C.** 500 kHz **D.** 50 kHz

Q4.08.1 MCQ 2008 1M

A 3-phase Voltage Source Inverter is operated in 180° conduction mode. Which one of the following statements is true?

A. Both pole-voltage and line-voltage will have 3rd harmonic components

B. Pole-voltage will have 3rd harmonic component but line-voltage will be free from 3rd harmonic

C. Line-voltage will have 3rd harmonic component but pole-voltage will be free from 3rd harmonic

D. Both pole-voltage and line-voltage will be free from 3rd harmonic components

Q4.08.2 MCQ 2008 2M

A single phase voltage source inverter is feeding a purely inductive load as shown in the figure.

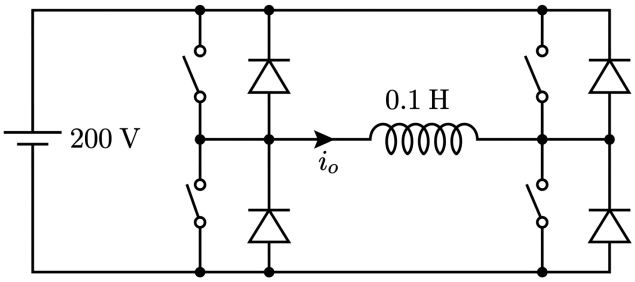


Fig. Q4.08.2

The inverter is operated at 50 Hz in 180° square wave mode. Assume that the load current does not have any dc component. The peak value of the inductor current i_o will be

A. 6.37 A **B.** 10 A **C.** 20 A **D.** 40 A

Q4.07.1 MCQ 2007 1M

"Six MOSFETs connected in a bridge configuration (having no other power device) MUST be operated as a Voltage Source Inverter (VSI)." This statement is:

A. True, because being majority carrier devices, MOSFETs are voltage driven.

B. True, because MOSFETs have inherently anti-parallel diodes.

C. False, because it can be operated both as a Current Source Inverter (CSI) and as a Voltage Source Inverter (VSI).

D. False, because MOSFETs can be operated as excellent constant current sources in the saturation region.

Q4.07.2 MCQ 2007 2M

A single-phase voltage source inverter is controlled in a single pulse-width modulated mode with a pulse width of 150° in each half cycle. Total harmonic distortion is defined as THD = $\frac{\sqrt{V_{rms}^2 - V_1^2}}{V_1} \times 100$, where V_1 is the rms value of the fundamental component of the output voltage. The THD of output ac voltage waveform is

A. 65.65% **B.** 48.42% **C.** 31.83% **D.** 30.49%

Q4.06.1 MCQ 2006 2M

A single-phase inverter is operated in PWM mode generating a single-pulse of width $2d$ in the centre of each half cycle as shown in figure. It is found that the output voltage is free from 5th harmonic for pulse width 144°. What will be percentage of 3rd harmonic present in the output voltage ($V_{o3}/V_{o1\,max}$) ?

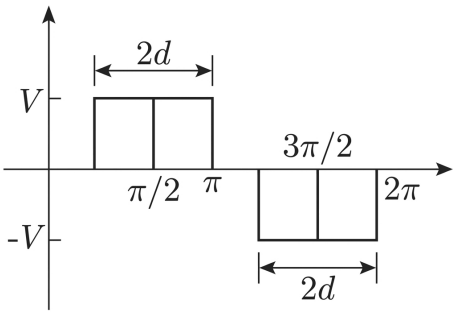


Fig. Q4.06.1

A. 0.0% **B.** 20.6% **C.** 31.7% **D.** 53.9%

Q4.05.1 MCQ 2005 1M

The output voltage waveform of a three-phase square-wave inverter contains

A. only even harmonics **B.** both odd and even harmonics
C. only odd harmonics **D.** only triple harmonics

Q4.03.1 MCQ 2003 2M

An inverter has a periodic output voltage with the output wave form as shown in figure.

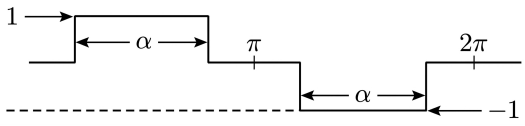


Fig. Q4.03.1

When the conduction angle $\alpha = 120°$, the rms fundamental component of the output voltage is

A. 0.78 V **B.** 1.10 V **C.** 0.90 V **D.** 1.27 V

Q4.03.2 MCQ 2003 2M

With reference to the output wave form given in above figure, the output of the converter will be free from 5th harmonic when

A. $\alpha = 72°$ **B.** $\alpha = 36°$
C. $\alpha = 150°$ **D.** $\alpha = 120°$

Q4.02.1 MCQ 2002 2M

The figure shows an inverter circuit with a dc source voltage V_s. The semiconductor switches of the inverter are operated in such a manner that the pole voltages v_{10} and v_{20} are as shown in Fig.Q4.02.1. What is the rms value of the pole-to-pole voltage v_{12}?

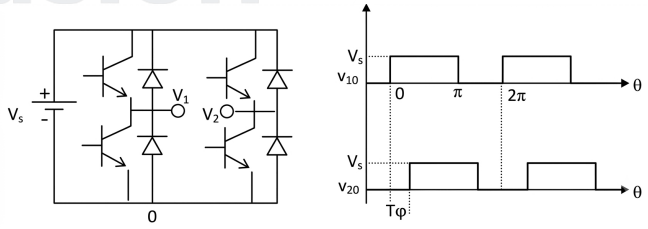


Fig. Q4.02.1

A. $\frac{V_s\phi}{\pi\sqrt{2}}$ **B.** $V_s\sqrt{\frac{\phi}{\pi}}$ **C.** $V_s\sqrt{\frac{\phi}{2\pi}}$ **D.** $\frac{V_s}{\pi}$

Q4.01.1 MCQ 2001 2M

A single-phase full-bridge voltage source inverter feeds a purely inductive load, a shown in Fig.Q4.01.1, where T_1, T_2, T_3, T_4 are power transistors and D_1, D_2, D_3, D_4 are feedback diodes. The inverter is operated in square-wave mode with a frequency of 50 Hz. If the average load

current is zero, what is the time duration of conduction of each feedback diode in a cycle?

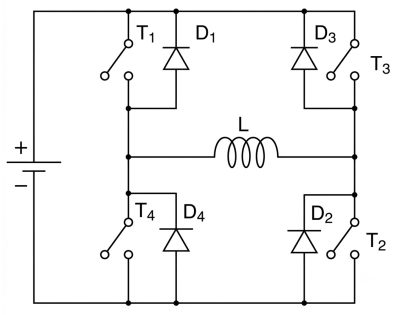


Fig. Q4.01.1

A. 5 msec **B.** 10 msec **C.** 20 msec **D.** 2.5 msec

Q4.00.1 MCQ 2000 2M

A three phase voltage source inverter supplies a purely inductive three phase load. Upon Fourier analysis, the output voltage waveform is found to have an h- th order harmonic of magnitude α_h times that of the fundamental frequency component ($\alpha_h < 1$). The load current would then have an h-th order harmonic of magnitude

A. zero

B. α_h times the fundamental frequency component

C. $h\alpha_h$ times the fundamental frequency component

D. α_h / h times the fundamental frequency component

Q4.00.2 MCQ 2000 1M

Triangular PWM control, when applied to a three phase, BJT based voltage source inverter, introduces

A. Low order harmonic voltages on the dc side

B. Very high order harmonic voltages on the dc side

C. Low order harmonic voltages on the ac side

D. Very high order harmonic voltage on the ac side

Q4.99.1 MCQ 1999 1M

A PWM switching scheme is used with a three phase inverter to

A. Reduce the total harmonic distortion with modest filtering

B. Minimize the load on the DC side

C. Increase the life of the batteries

D. Reduce low order harmonics and increase high order harmonics

Q4.97.1 NAT 1997 5M

A single-phase bridge inverter is fed from a 200V d.c. supply and is operated at 50Hz. It is connected to a load having a resistance of 20 ohms and an inductance of 0.2H. Draw the load current waveform in the steady state indicating the peak values.

Q4.95.1 NAT 1995 1M

A single phase inverter with square wave voltage will have in its output waveform a fifth harmonic component equal to _____ percentage of the fundamental.

Q4.95.2 MCQ 1995 1M

An inverter capable of supplying a balanced three-phase variable voltage variable frequency output is feeding a three-phase induction motor rated for 50Hz and 440V. The stator winding resistances of the motor are negligibly small. During starting, the current inrush can be avoided without sacrificing the starting torque by suitably applying:

A. Low voltage at rated frequency

B. Low voltage keeping the V/f ratio constant

C. Rated voltage at low frequency

D. Rated voltage at rated frequency

Answer Key

Note: * descriptive/evidence-based, no single definitive answer.

Q4.26.1 B	Q4.25.1 A	Q4.24.1 50 to 50
Q4.22.1 3183	Q4.22.2 0.46 to 0.48	Q4.21.1 50.00 to 52.00
Q4.20.1 19.90 to 20.20	Q4.19.1 111 to 115	Q4.17.1 197 to 199
Q4.17.2 9 to 9	Q4.16.1 9.9 to 10.1	Q4.16.2 A
Q4.15.1 62.75	Q4.14.1 76.5 to 78.0	Q4.14.2 D
Q4.13.1 D	Q4.13.2 D	Q4.12.1 B

Q4.12.2 D	Q4.11.1 A	Q4.09.1 B
Q4.08.1 B	Q4.08.2 B	Q4.07.1 B
Q4.07.2 C	Q4.06.1 B	Q4.05.1 C
Q4.03.1 A	Q4.03.2 A	Q4.02.1 B
Q4.01.1 A	Q4.00.1 D	Q4.00.2 D
Q4.99.1 D	Q4.97.1 *	Q4.95.1 20
Q4.95.2 B		

PrepFusion

UNIT

V

Miscellaneous Topics

Topics

1. AC Voltage Regulator

Unit Snapshot*

Stream: EE	**Questions:** 5	**MCQ:** 5	**MSQ:** 0	**NAT:** 0
1M: 2	**2M:** 3	**Descriptive:** 0	**Avg Weightage** ~1M	**Range:** 1M(2016)–2M(2024)

**See preface for how Avg Weightage and Range are calculated.*

Q5.24.1 MCQ 2024 2M

A single-phase TRIAC-based AC voltage controller feeds a series RL load. The input AC supply is 230 V, 50 Hz. The values of R and L are 10 Ω and 18.37 mH, respectively. The minimum triggering angle of the TRIAC to obtain controllable output voltage is

A. 15° **B.** 30° **C.** 45° **D.** 60°

Q5.16.1 MCQ 2016 1M

In a constant V/f induction motor drive, the slip at the maximum torque

A. is directly proportional to the synchronous speed.

B. remains constant with respect to the synchronous speed.

C. has an inverse relation with the synchronous speed.

D. has no relation with the synchronous speed.

Q5.14.1 MCQ 2014 2M

A single-phase SCR-based AC regulator is feeding power to a load consisting of 5 Ω resistance and 16 mH inductance. The input supply is 230 V, 50 Hz AC. The maximum firing angle at which the voltage across the device becomes zero all throughout and the rms value of current through SCR, under this operating condition, are _____.

A. 30° and 46 A **B.** 30° and 23 A
C. 45° and 23 A **D.** 45° and 32 A

Q5.08.1 MCQ 2008 1M

In the single phase voltage controller circuit shown in the figure, for what range of triggering angle (α), the output voltage (v_o) is not controllable?

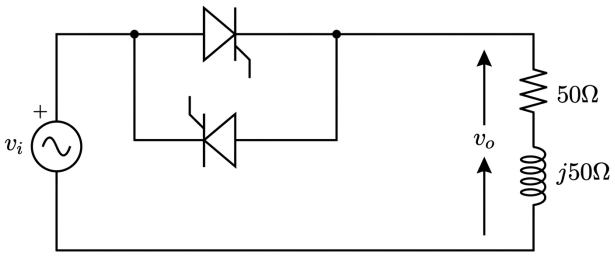
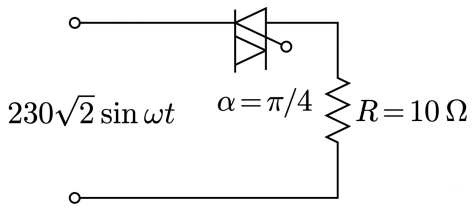


Fig. Q5.08.1

A. $0° < \alpha < 45°$ **B.** $45° < \alpha < 135°$
C. $90° < \alpha < 180°$ **D.** $135° < \alpha < 180°$

Q5.04.1 MCQ 2004 2M

The TRIAC circuit shown in the figure controls the AC output power to the resistive load. The peak power dissipation in the load is



A. 3968 W **B.** 5290 W **C.** 7935 W **D.** 10580 W

Answer Key

Q5.24.1 **B**	Q5.16.1 **C**	Q5.14.1 **C**
Q5.08.1 **A**	Q5.04.1 **D**	

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Engineering Mathematics	Linear Algebra, Calculus, Differential Equations, Complex Variables, Probability & Statistics, Numerical Methods, Transform Theory and Vector Calculus — all branches
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